# Computer Network Project
# On
# Educational Network for School with Segmented User Roles

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**LAB PROJECT REPORT**

This Report Presented in Partial Fulfillment of the course

**CSE322: Computer Network Lab in the Department of Computer Science and Engineering**


**DAFFODIL INTERNATIONAL UNIVERSITY**

**Dhaka, Bangladesh**


**11/Auguest/2025**

## DECLARATION

We hereby declare that this lab project has been done by us under the supervision of **Partha Dip Sarkar,** Lecturer, Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere as lab projects.

**Submitted To:**

_____

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# COURSE & PROGRAM OUTCOME

The following course have course outcomes as following:.

Table 1: Course Outcome Statements

| CO's | Statements |
|---|---|
| CO1 | Define and relate network components, such as routers, switches, DHCP, DNS, and NAT |
| CO2 | Apply IP subnetting and address planning to segment a network into multiple user groups. |
| CO3 | Analyze network topologies and routing tables for optimizing communication and security. |
| CO4 | Design and implement a secure and scalable school network applying standard networking tools. |

Table 2: Mapping of CO, PO, Blooms, KP and CEP

| CO | PO | Blooms | KP | CEP |
|---|---|---|---|---|
| CO1 | PO1 | C1, C2 | KP3 | EP1, EP3 |
| CO2 | PO2 | C2 | KP3 | EP1, EP3 |
| CO3 | PO3 | C4, A1 | KP3 | EP1, EP2 |
| CO4 | PO3 | C3, C6, A3, P3 | KP4 | EP1, EP3 |

The mapping justification of this table is provided in section **4.3.1**, **4.3.2** and **4.3.3**.

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# Chapter 1

# Introduction

## 1.1 Introduction

In modern educational institutions, seamless and secure network connectivity has become essential for administrative tasks, academic activities, and guest services. Schools must cater to different categories of users—staff, students, and guests—each with distinct access requirements and security needs. However, without proper segmentation, unauthorized access to sensitive resources or interference between groups can occur, leading to performance issues and security vulnerabilities.

This project focuses on designing and implementing a **secure and segmented school network** with dedicated subnets for staff, students, and guests, integrated with internet access through multiple simulated Internet Service Providers (ISPs). The network ensures role-based access control, efficient IP allocation, and smooth service availability through static routing, DHCP, DNS, HTTP servers, wireless access points, and NAT configuration.

The design also simulates real-world networking principles such as internal/external DNS resolution, separation of traffic through SSIDs, and controlled communication between subnets. This solution provides both functionality and security while reflecting practical implementations in real schools.

## 1.2 Motivation

The growing dependency on digital resources in education has amplified the importance of secure, reliable, and well-structured networks. Schools not only require access to educational websites and internal resources but also need to prevent unauthorized access and maintain data privacy.

In many educational institutions, inadequate network segmentation results in:

- Security breaches due to open access between users
- Bandwidth congestion caused by uncontrolled guest usage
- Administrative inefficiencies in IP management and routing

By implementing a segmented network with proper routing, DHCP, DNS, and access control, this project ensures:

- Enhanced **security** for sensitive staff data
- **Uninterrupted internet** access for students
- **Controlled guest access** without affecting academic resources

This project also serves as a hands-on learning opportunity to apply networking concepts such as subnetting, NAT, and static routing in a realistic educational environment.

## 1.3 Objectives

The objectives of this project are:

1. **Design and implement** a segmented network for staff, students, and guests using subnetting.
2. **Configure DHCP** for automated IP allocation across all groups.
3. **Implement static routing** to control communication between subnets and ensure internet access.
4. **Deploy internal and external DNS** for name resolution.
5. **Host an internal HTTP server** for school use and simulate access to external educational sites.
6. **Provide wireless connectivity** for students and guests with separate SSIDs.
7. **Set up NAT** to allow internal hosts to reach the simulated internet.
8. **Integrate multiple ISP routers** to simulate real-world internet infrastructure.

## 1.4 Feasibility Study

Various educational networks have been implemented using FLAN segmentation, wireless SSID separation, and centralized DHCP services. For example, **W3Schools Campus Network Design** and **Cisco Packet Tracer-based educational labs** have demonstrated similar configurations for practical training. In real-world scenarios, many schools implement **role-based access control** to separate administrative and academic traffic, often using managed switches and firewall rules.

Existing studies show that:

- DHCP and DNS integration improve network scalability and usability.
- Segmentation improves security by isolating traffic.
- Static routing and NAT ensure controlled communication with external networks.

Our approach builds upon these concepts but adapts them into a fully simulated environment within **Cisco Packet Tracer**, ensuring that multiple ISPs, internal/external DNS, and HTTP services are integrated in one cohesive setup.

## 1.5 Gap Analysis

While existing network designs in schools provide basic connectivity, many lack:

- Simulated **multi-ISP** environments for redundancy testing
- Fully integrated **internal + external DNS** resolution in a single topology

- **Guest network separation** at both wired and wireless levels
- Implementation of **multiple HTTP servers** to simulate real-world educational sites
- Emphasis on **static routing** in place of dynamic protocols for better teaching control

This project addresses these gaps by designing a **comprehensive, fully segmented, and simulated educational network** that can serve as both a functional prototype and a teaching model.

## 1.6 Project Outcome

Upon completion, the project is expected to deliver:

- A **secure school network topology** with staff, student, and guest subnets
- **Automated IP management** through DHCP for each subnet
- **Role-based access control** via static routing rules
- **Working internal DNS server** and resolution to external DNS sites
- **Functional internal HTTP server** hosting school resources
- **Wireless access points** with separate SSIDs for students and guests
- **Simulated internet environment** with multiple ISP routers and HTTP servers
- A scalable, testable, and **educational reference network** for future implementations

# Chapter 2

# Proposed Methodology/Architecture

The network's technical design process is described in this chapter. It comprises the system design, requirement analysis, and implementation strategy in detail.

## 2.1 Requirement Analysis & Design Specification

### 2.1.1 Overview

The proposed network aims to provide **secure and segmented internet access** for staff, students, and guests within a school environment. The design focuses on **role-based access control**, **controlled inter-subnet communication**, and **efficient resource utilization** through the implementation of DHCP, DNS, NAT, and static routing.
Simulation is carried out in **Cisco Packet Tracer**, which allows for accurate modeling of routers, switches, access points, and servers.

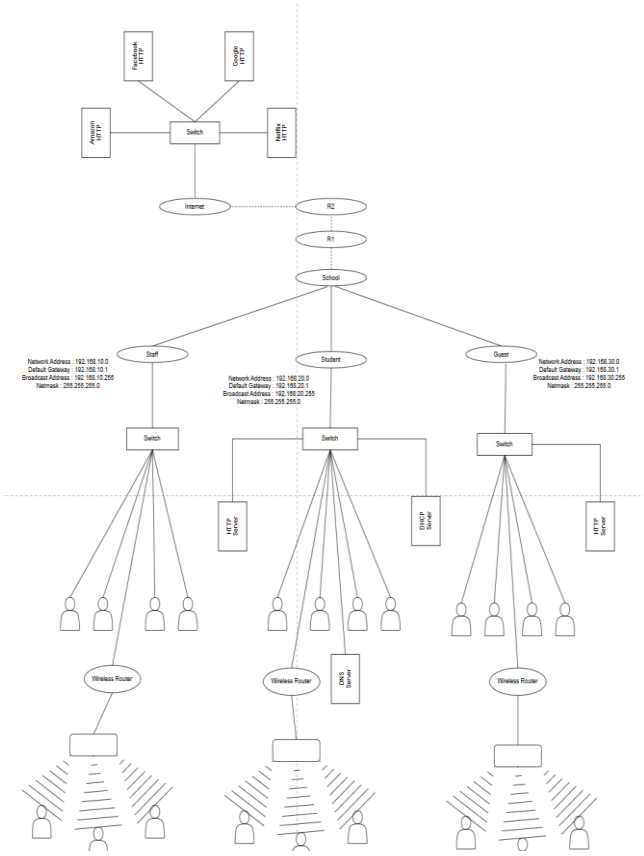
The design ensures:

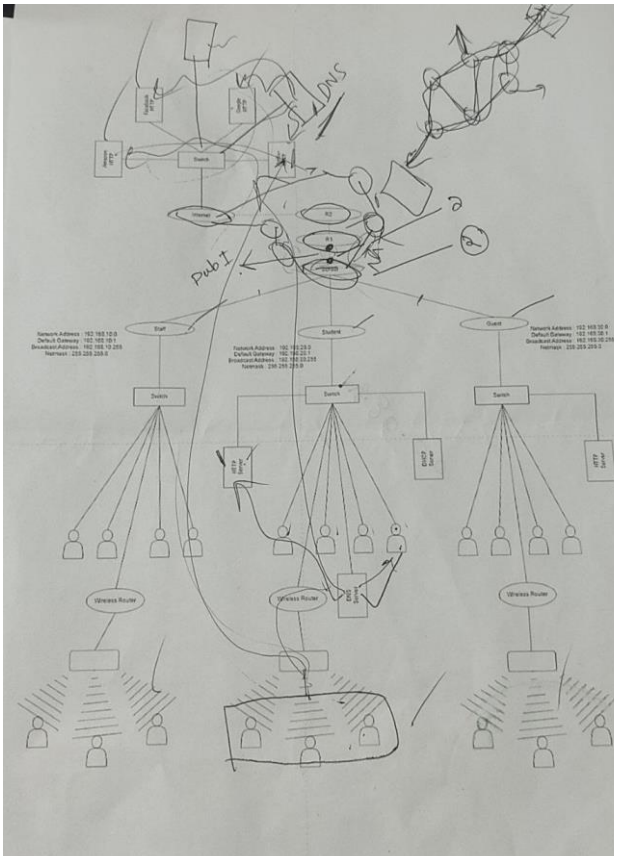
- Segmentation of traffic into **three main subnets**: Staff, Student, and Guest.
- Separate **wired and wireless** access for staff, students, guests.
- Integration of **internal and external HTTP servers**.
- Proper **DNS resolution** for both internal and internet-based resources.
- **Static routing** to manage traffic paths securely and predictably.
- **NAT configuration** to allow private IP devices to access external networks.

### 2.1.2 Proposed Methodology/ System Design

The system design follows a **layered approach**:

1. **Planning & Approving**
    - Make a Paper plan Diagram according to our knowledge on our project criteria
    - Getting Sir approve and suggestion to execute the project




**Proposed By Us** **Fixed By Sir**

## Subnetting & IP Planning

- The IP block 172.20.0.0/16 is subnetted using **FLSM** into multiple /24 networks.
- **9 subnets** are assigned to various network segments, ensuring clear isolation and easier troubleshooting.

2. **Assigning & Documentation**
   - We then configure Routers, end devices, HTTP, DNS, DHCP servers.
   - We use proper Documentation For better and easy understanding and easy reusability
3. **Core Routing Layer**
   - The main school router connects to multiple ISP routers through NAT for internet simulation.
   - Static routing is implemented on all routers to ensure controlled communication.

- We have also implemented Default routing

4. **Access Layer**
    - Switches connect end devices (PCs, Smartphone, servers) within each subnet.
    - Wireless routers are configured for staff, student and guest access with **unique SSIDs** and **WPA security**.
5. **Server Layer**
    - DHCP servers assign IP addresses dynamically for each subnet.
    - DNS server resolves both internal and external domain names.
    - HTTP servers provide school-specific resources internally and simulate educational websites externally.
6. **Security Layer**
    - Guests Wireless Device cannot access Internal network resources only have access to Internet.

NAT (Static and Dynamic) ensures internal devices can safely access the internet while keeping private IPs hidden.

### 2.1.3 UI Design

#### 1. Entire Image

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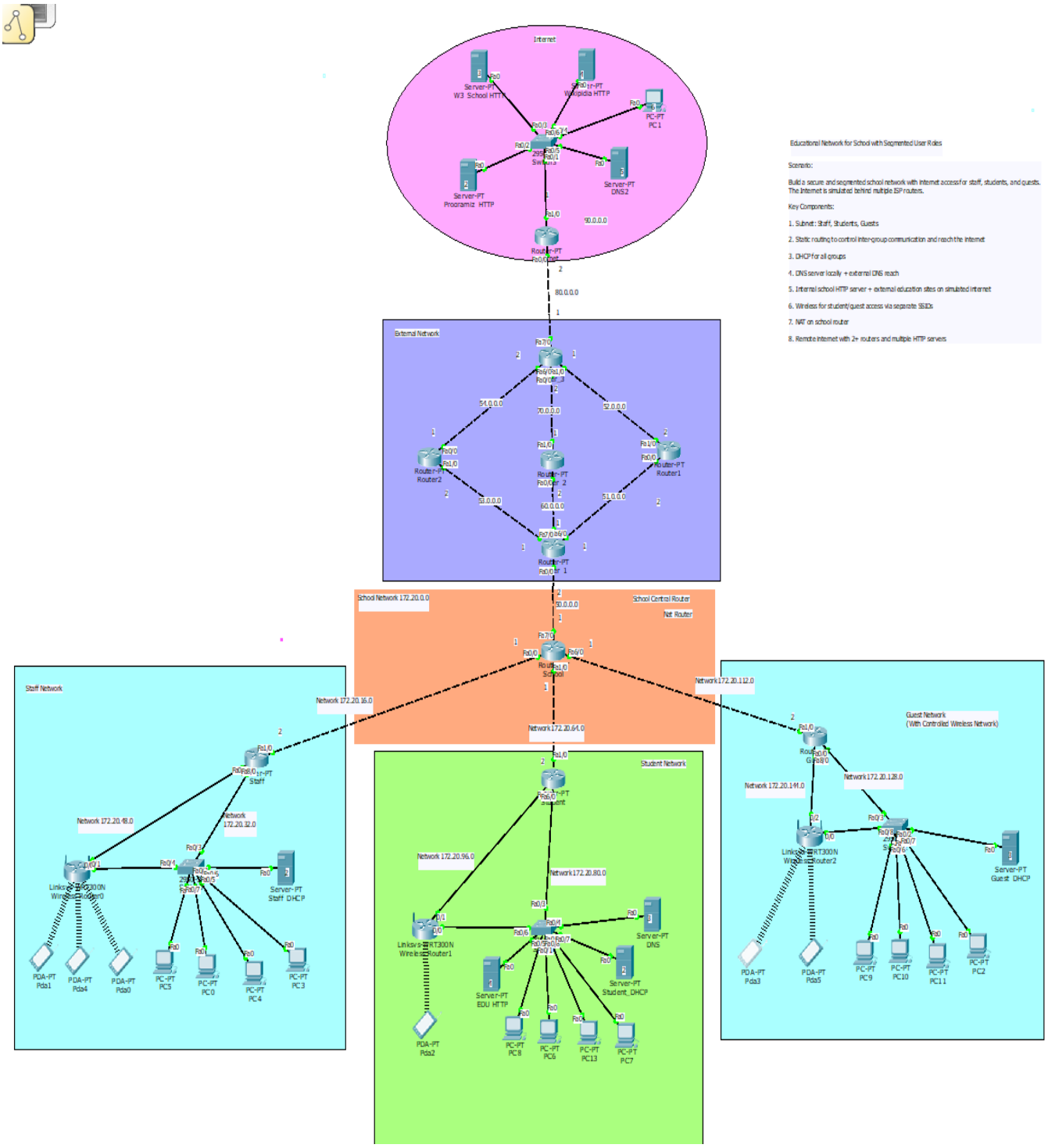


**Fig 1:** School Network with Secure Internet Access design

## 2.Subnetted Network

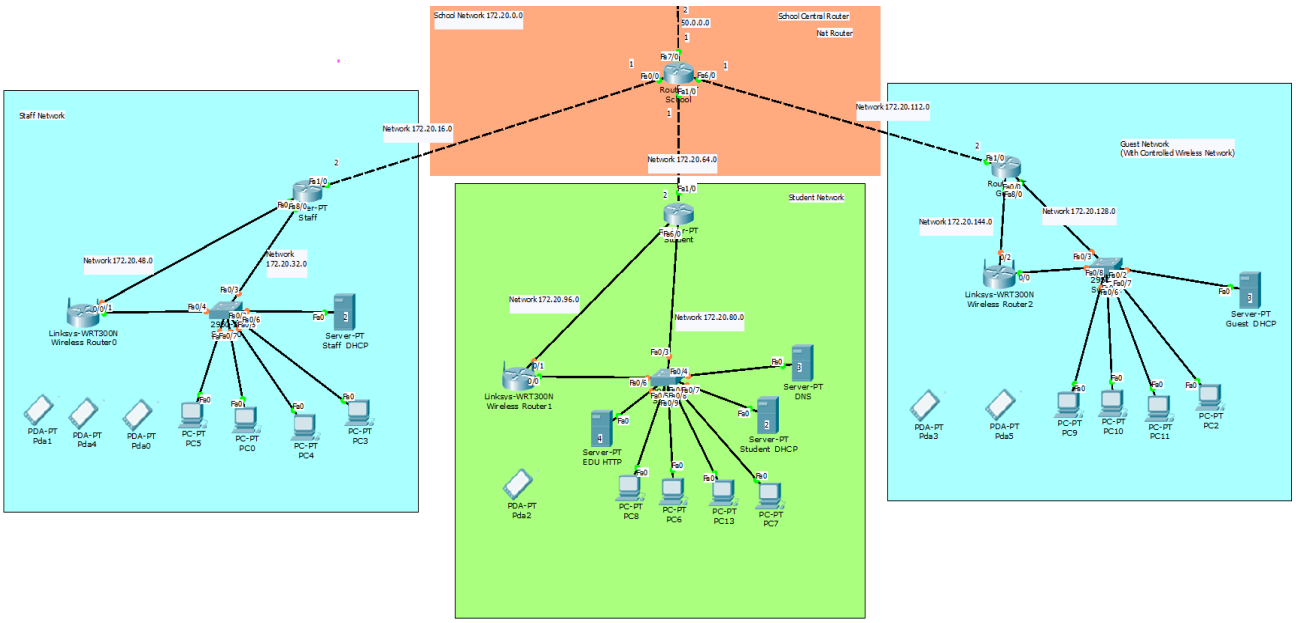


Fig 2: 9 Used Subnetted Network

**3.**

## 3.External Network

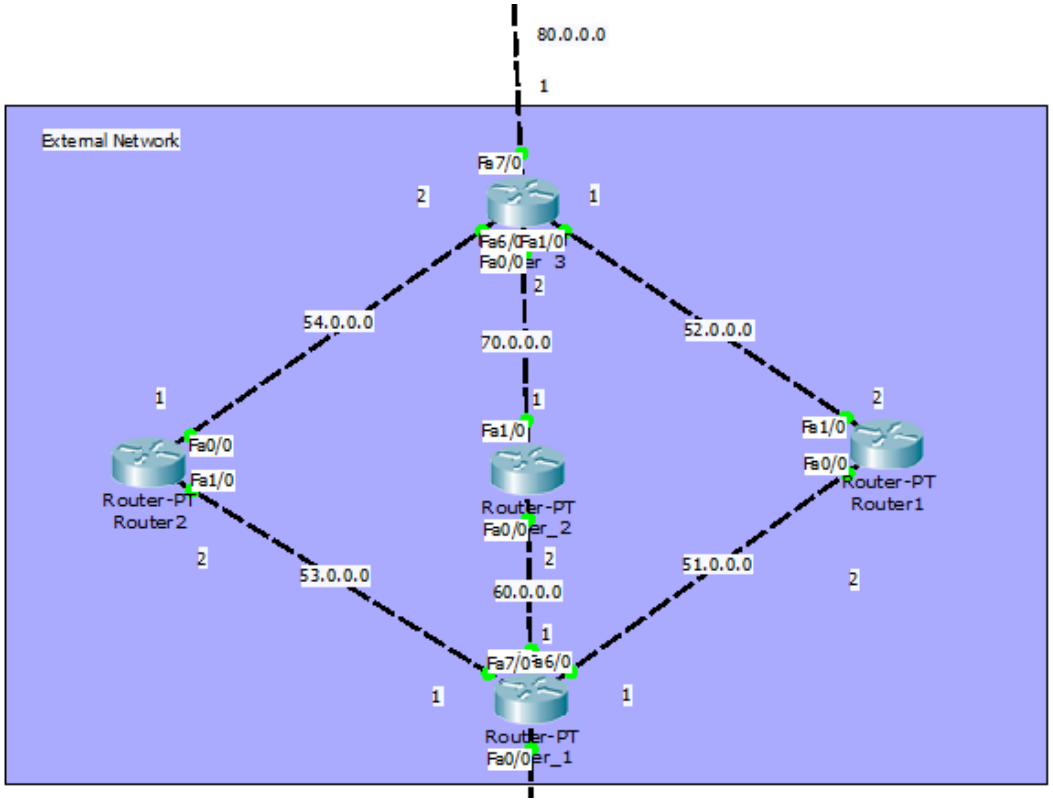


Fig 3: Rest of the internet 1

**4.Internet**

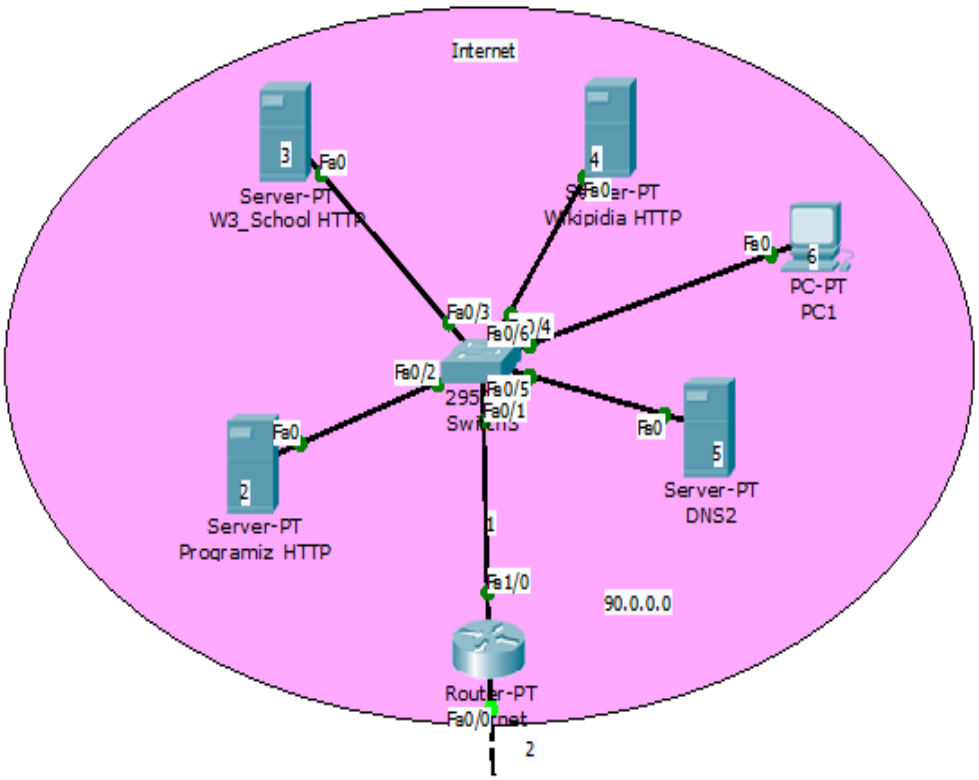


**Fig 4:** Rest of the internet 2

## 2.2 Overall Project Plan

The implementation is carried out in the following phases:

1. **Planning Phase**
    - Design a School Network with pen and paper.
    - Draw down using Diagram.ai.
    - Consult with Sir and Implement in Cisco Packet Tracer
    - Planning network and Subnets for network.
    - Allocate IP ranges for each subnet with the help of AI.
    - Assigning IP in the planed Destination with proper documentation for Easy configuration.

2. **Configuration Phase**
    - Configure routers with static routes.
    - Set up DHCP, DNS, and HTTP servers of internal and external network.
    - Configure wireless routers and SSIDs.
    - Implement NAT in CLI of School Router.
3. **Testing Phase**
    - Verify connectivity between all devices.
    - Test internal and external DNS resolution.
    - Check HTTP access to internal and external servers.
    - Ping out different IP in the command section to check reply.
4. **Documentation Phase**
    - Record configurations, subnet details, and network diagrams.
    - Compile project report with screenshots and analysis.

# Chapter 3

# Implementation and Results

Here we described the implementation of the proposed network design in Cisco Packet Tracer and presents the performance analysis and key results obtained from testing.

## 3.1 Implementation

### 3.1.1. Subnet Configuration

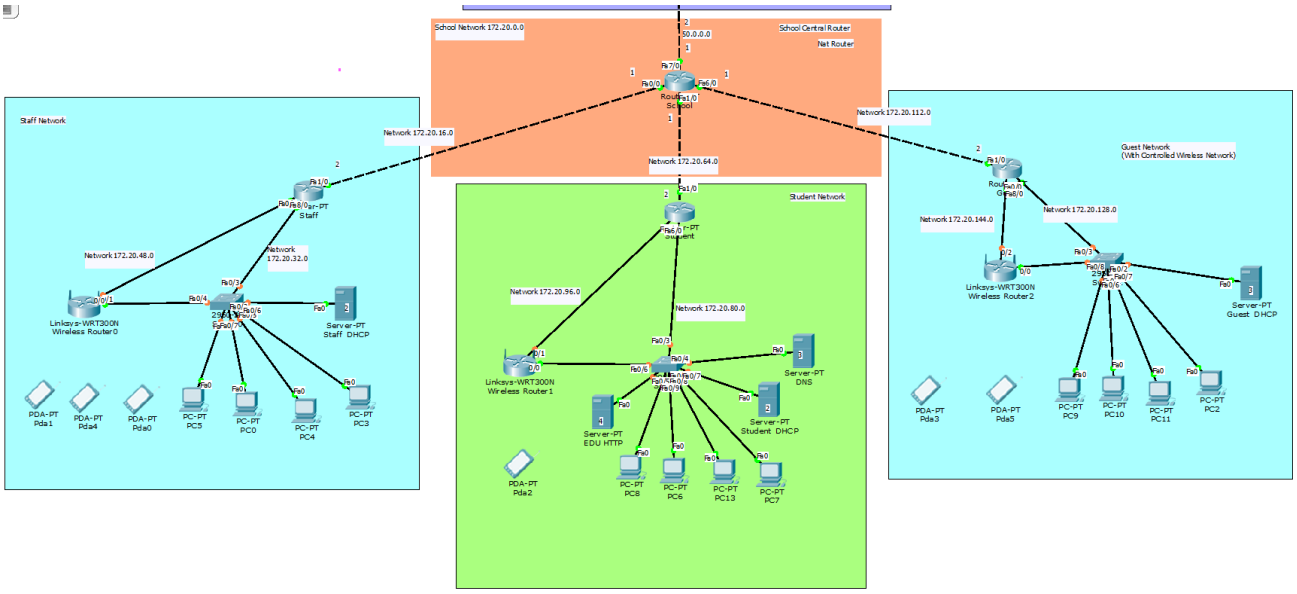


Fig 1: Subnetted network

For subnetting Network IP 172.20.0.0 is divided into 255 groups using FLSM rules. The subnet mask of each subnet is 255.255.240.0. From here 9 subnets is used in the network. The details are given below:

| Subnet | Router | Used Port | Roles | Device Included |
|---|---|---|---|---|
| 172.20.16.0 | School | Fa0/0 | Staff Network | Smart phones, Pcs, DHCP server, http srver, Switch, DNS Server |

| | | | | |
|---|---|---|---|---|
| 172.20.48.0 | Staff | Fa8/0 | Staff Wifi | Smart phones(wireless Device) |
| 172.20.32.0 | Staff | Fa0/0 | Staff Wired | Pcs,Servers |
| 172.20.64.0 | School | Fa1/0 | Student Network | Smart phones, Pcs, DHCP server, http srver, Switch, DNS Server |
| 172.20.96.0 | Student | Fa6/0 | Student Wifi | Smart phones(wireless Device) |
| 172.20.80.0 | Student | Fa0/0 | Student Wired | Pcs,Servers |
| 172.20.112.0 | School | Fa6/0 | Guest Network | Smart phones, Pcs, DHCP server, http srver, Switch, DNS Server |
| 172.20.144.0 | Guest | Fa8/0 | Guest Wifi | Smart phones(wireless Device) |
| 172.20.128.0 | Guest | Fa0/0 | Guest Wired | Pcs,Servers |

### 3.1.2 Wireless Router Configuration

There is a wireless router which is connected with staff, student, guest router each. They provide connection with internet and internal network using different SSID's and password. Also **Controlled wireless Guest router** with prohibation in accessing Internal network.

Here the details of wireless router

| Name | WAN | LAN | DHCP Range |
|---|---|---|---|
| Wireless Router 0 | 172.20.16.0 | 172.20.48.0 | 172.20.48.10-109 |
| Wireless Router 1 | 172.20.64.0 | 172.20.96.0 | 172.20.96.10-109 |
| Wireless Router 2 | 172.20.112.0 | 172.20.144.0 | 172.20.144.1-50 |

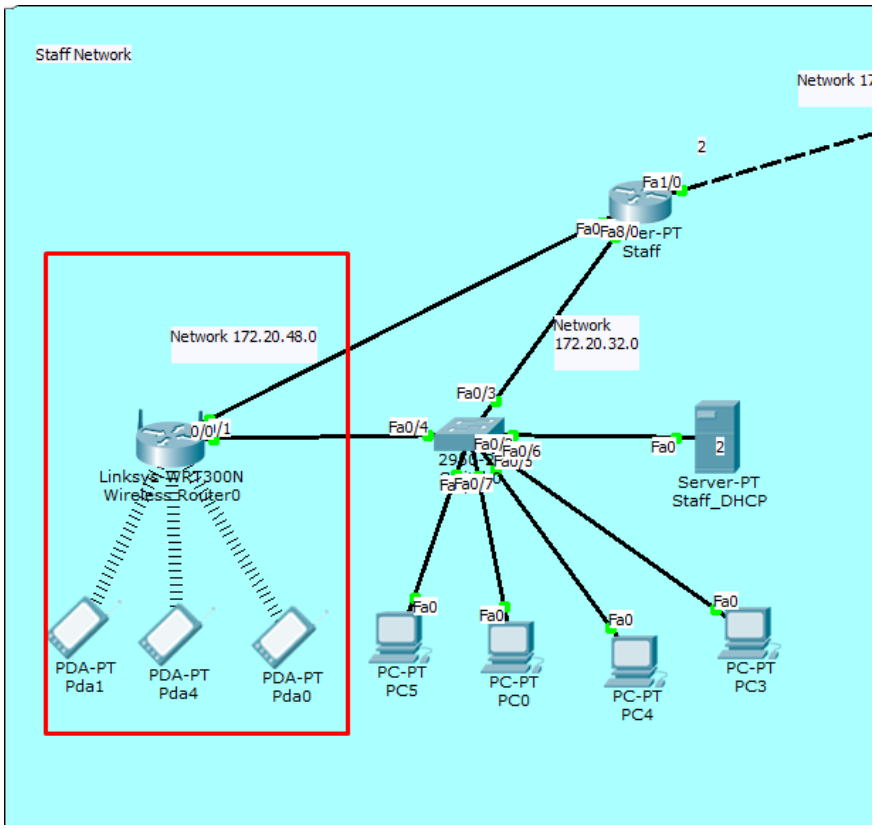


Fig 1: Guest Wireless Access

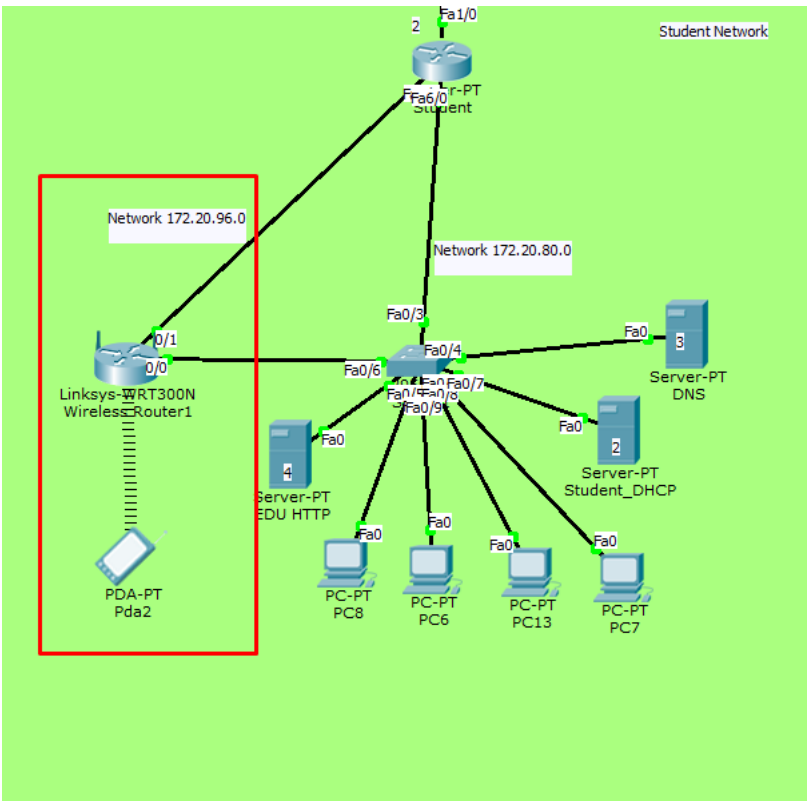


Fig 2: Student Wireless Access

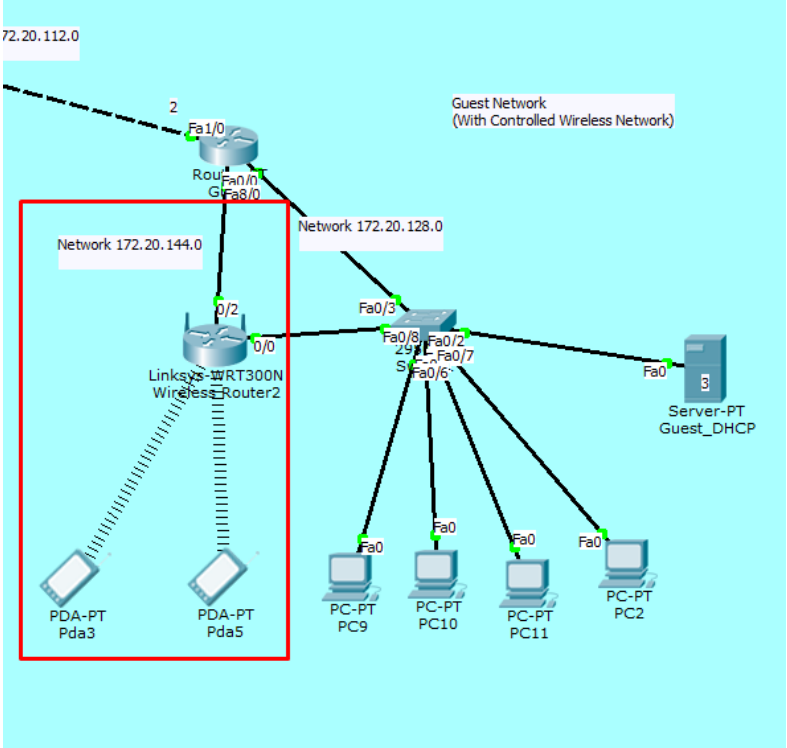


Fig 3: Guest Wireless Access

### 3.1.3 Static Routing Configuration

Static routing involves manually configuring routes in the routing table. It is suitable for small, stable networks where routes do not change frequently.

There are **total 4 router inside private network 6 router outside private network** where we use static routing configuration for communications. For static routing we first go to the selected router and go to config tab. From there we have to select static and have to assign required network IP, its subnetting mask and next hop.

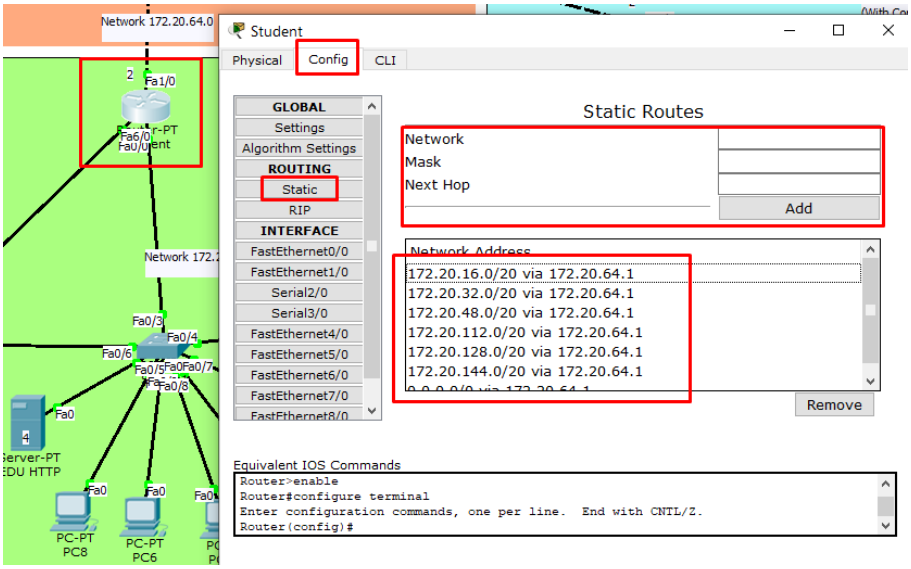


Fig 1: Configuration of Static Routing in Student Router

If we want to check the route of router, we just simply go to CLI tab and enable it. Then we have to write "**show ip route**".

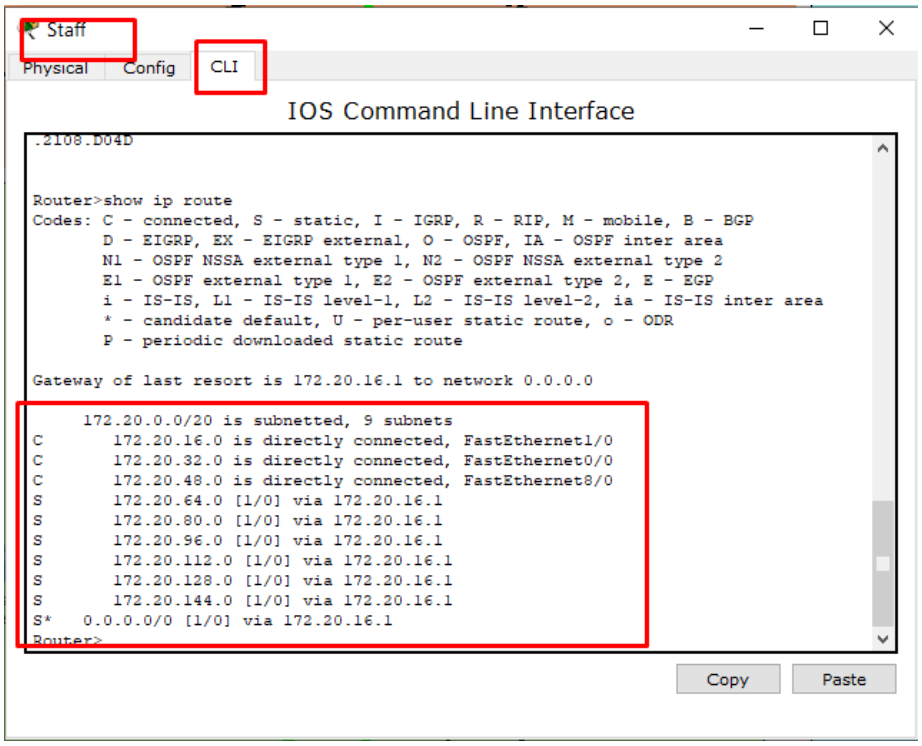


Fig 2: Static Routing in Staff Router

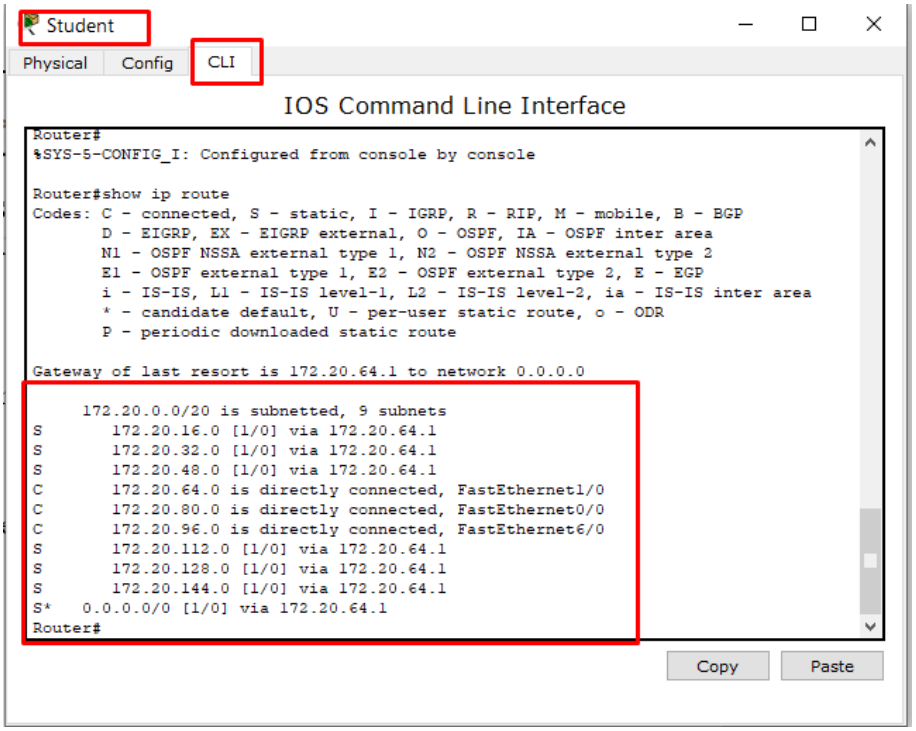


Fig 3: Static Routing in Student Router

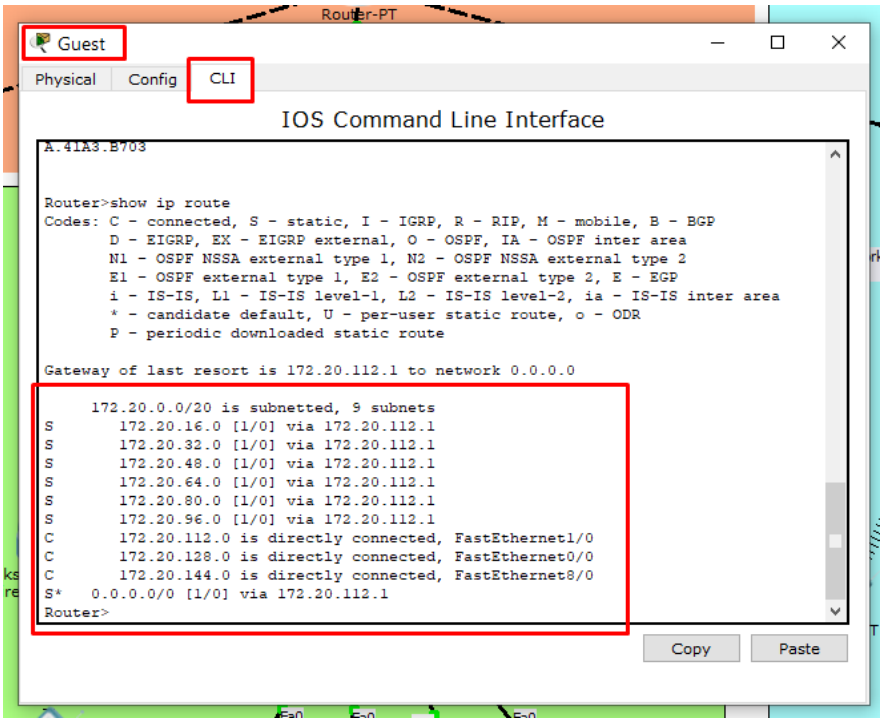


Fig 4: Static Routing in Guest Router

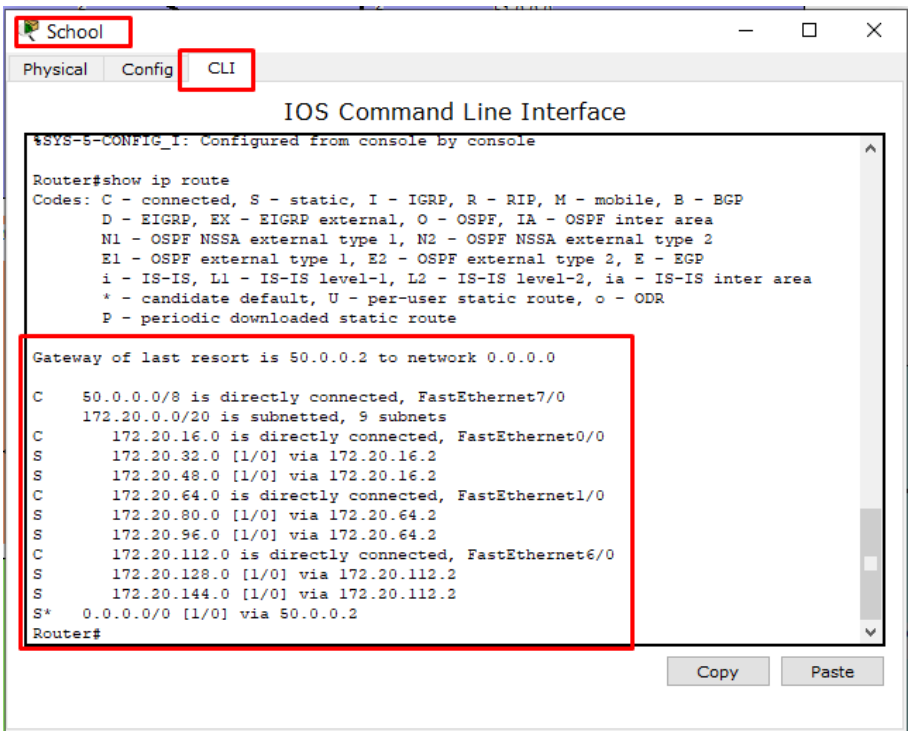


Fig 5: Static Routing in Guest Router

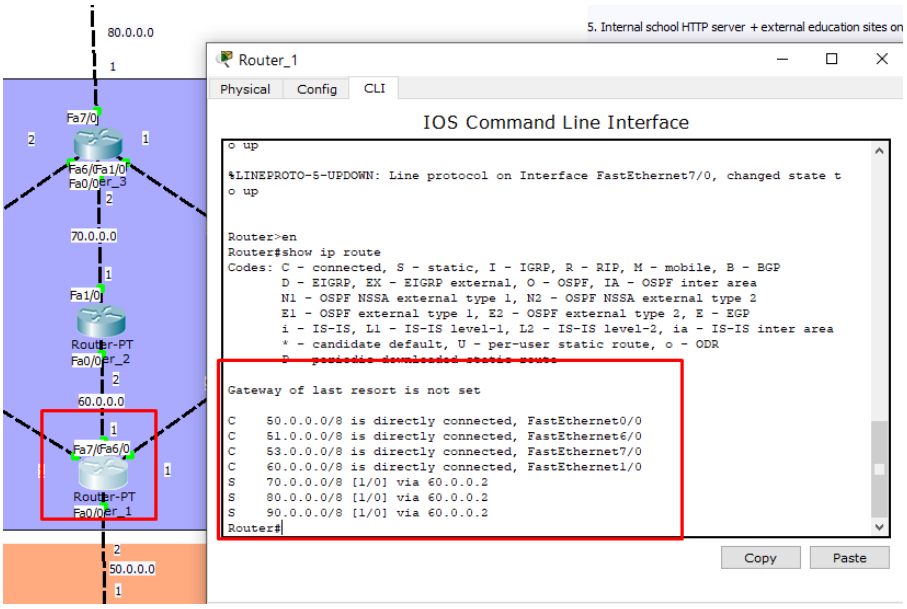


Fig 6: Static Routing in External Router

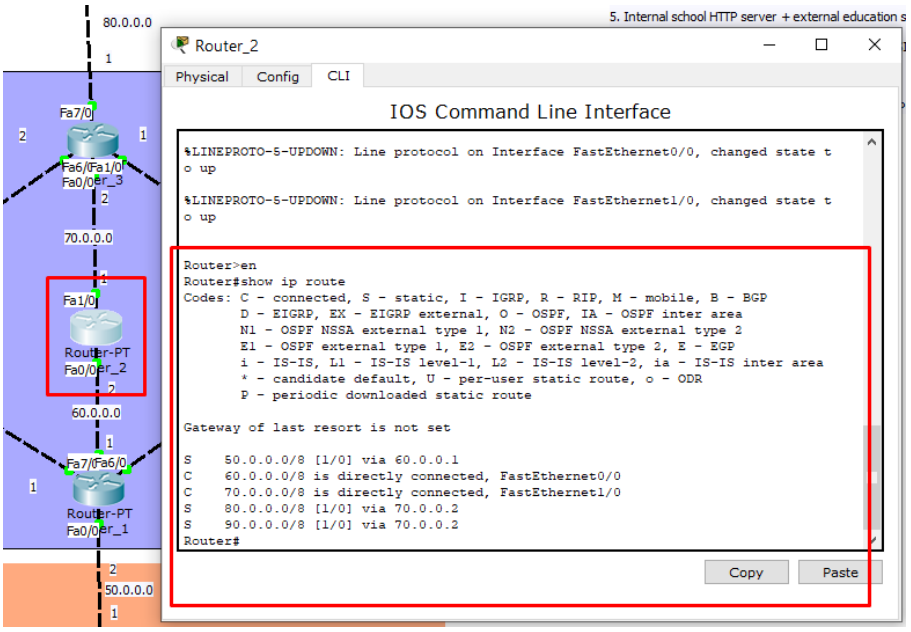


Fig 7: Static Routing in External Router

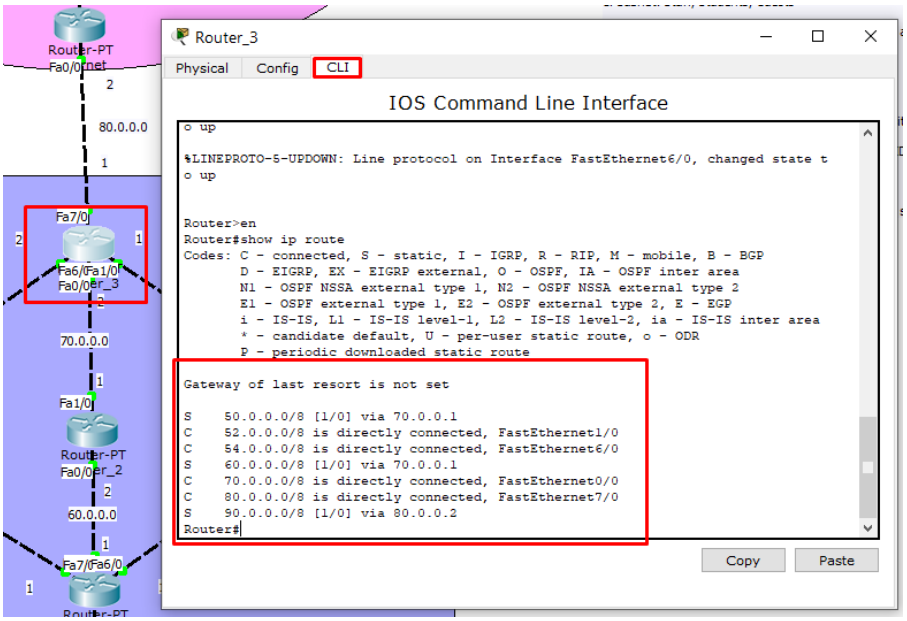


Fig 8: Static Routing in External Router

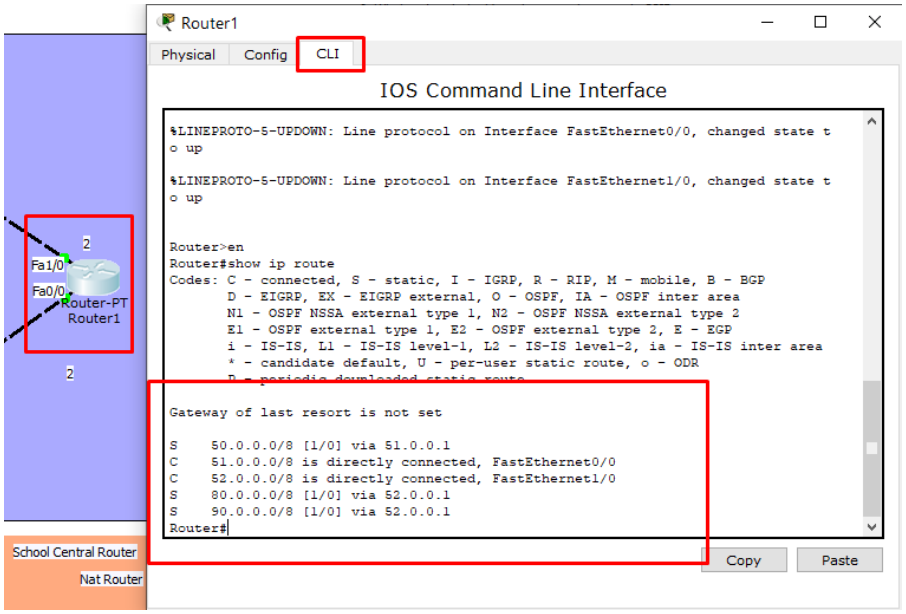


Fig 9: Static Routing in External Router

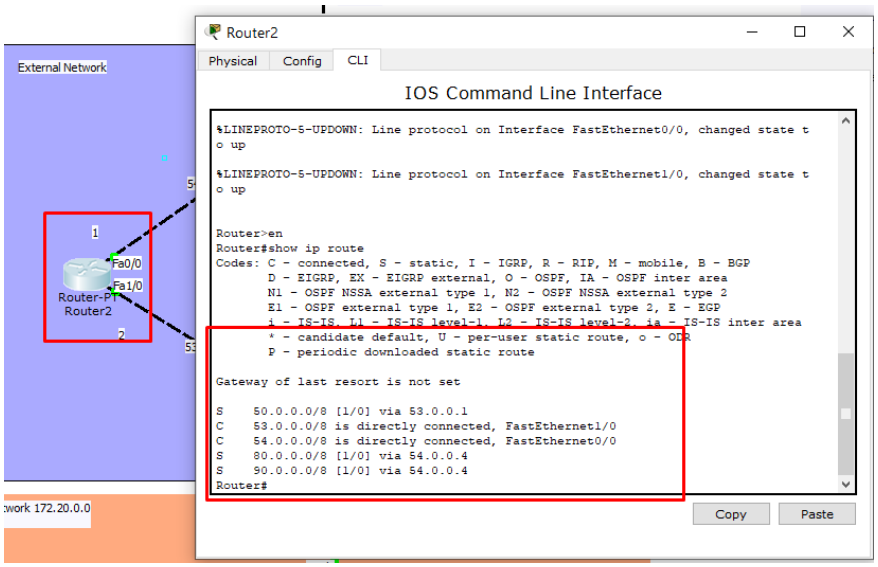


Fig 10: Static Routing in External Router

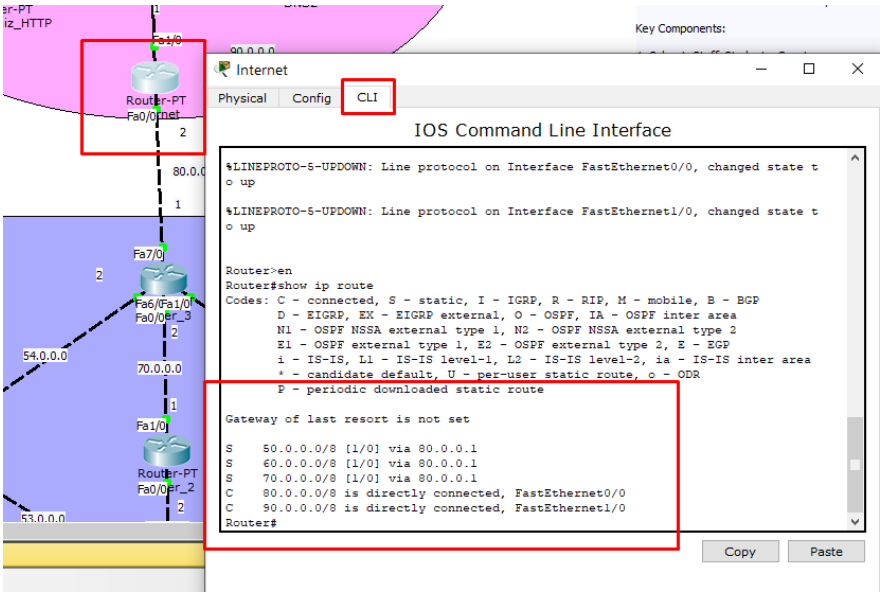


Fig 11: Static Routing in Internet Router

### 3.1.4 DHCP SETUP

DHCP (Dynamic Host Configuration Protocol) is a networking protocol that automatically assigns IP addresses and other configuration settings to devices on a network.

There is total of three DHCP servers in the network. One is for staff Pcs, one for student PCs and one for guest PCs.We use this to ease our device ip configuration.

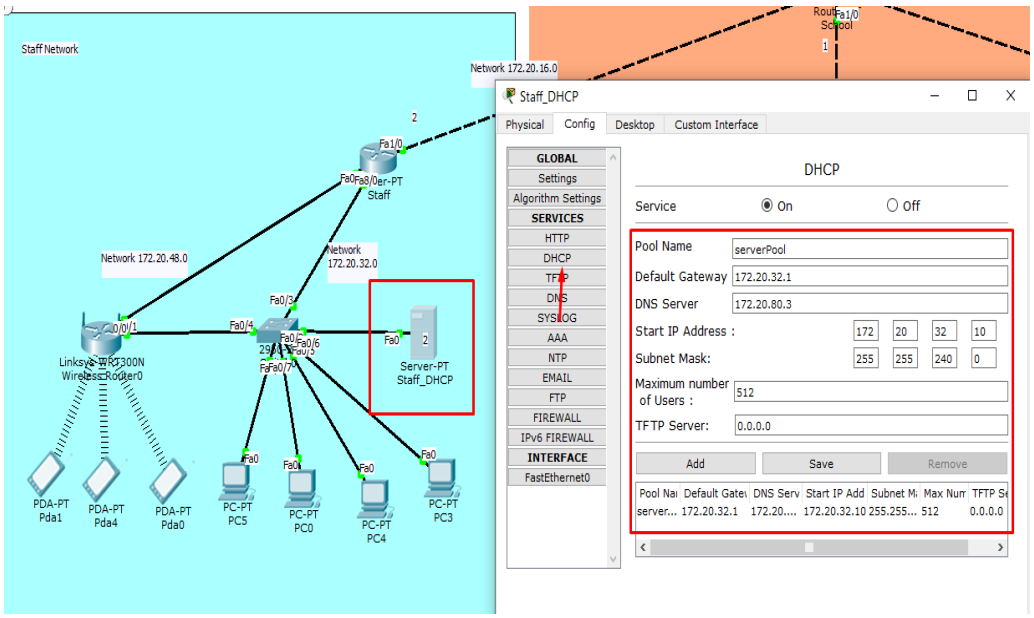


Fig 1: Staff DHCP Server

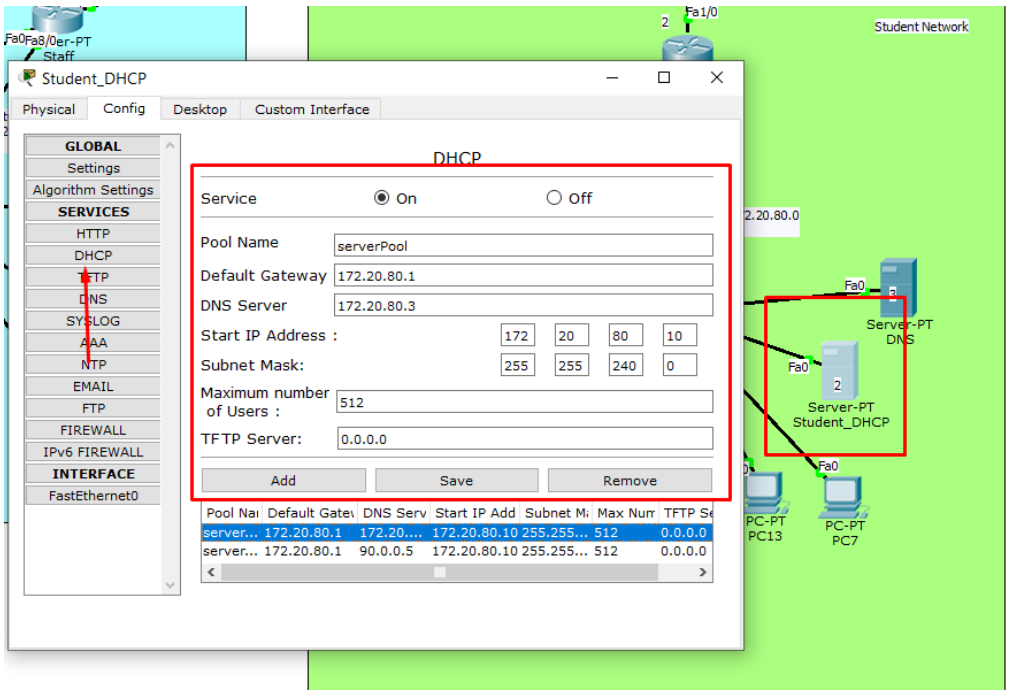


Fig 2: Student DHCP Server

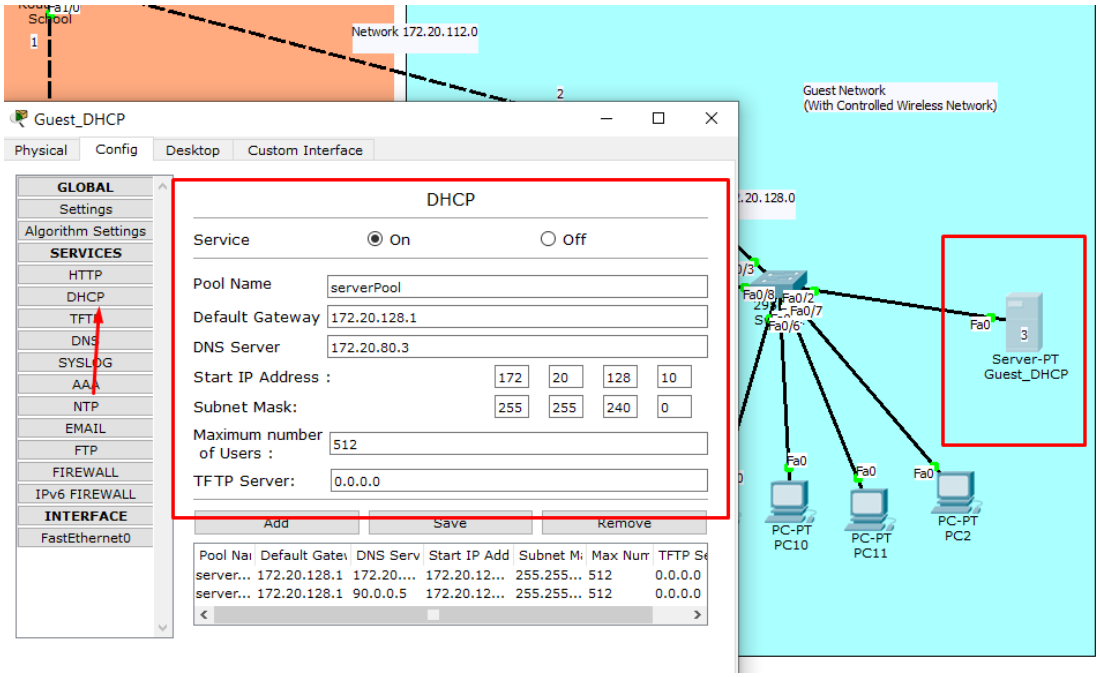


Fig 3: Guest DHCP Server

### 3.1.5 DNS Server Setup

The Domain Name System (DNS) is a foundational component of the internet that translates human-readable domain names (e.g., www.example.com) into machine-readable IP addresses (e.g., 192.0.2.1).
There is one DNS server in the internal network which can work on both internal and external HTTPs servers.

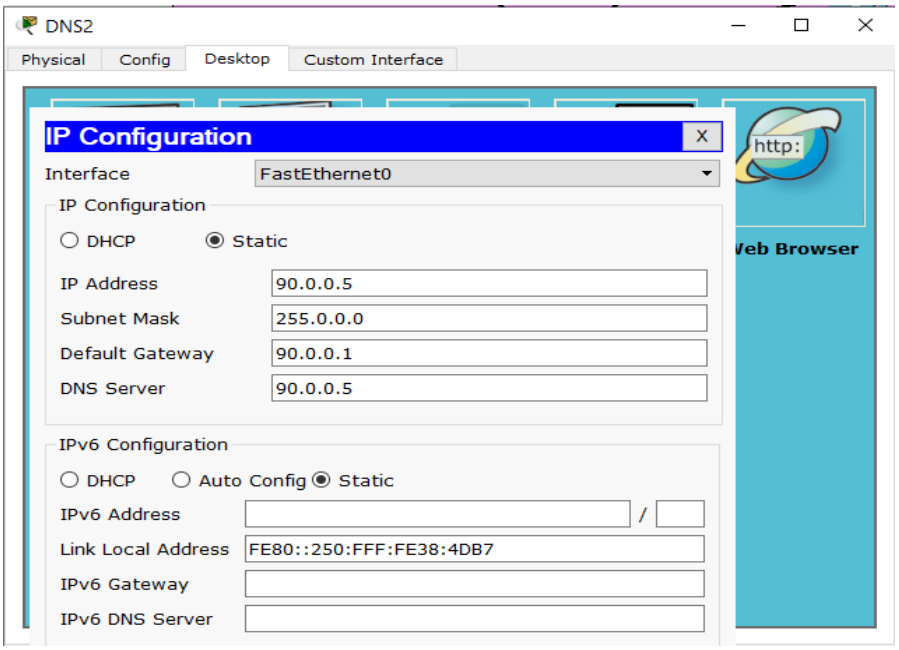


Fig 1: External DNS server

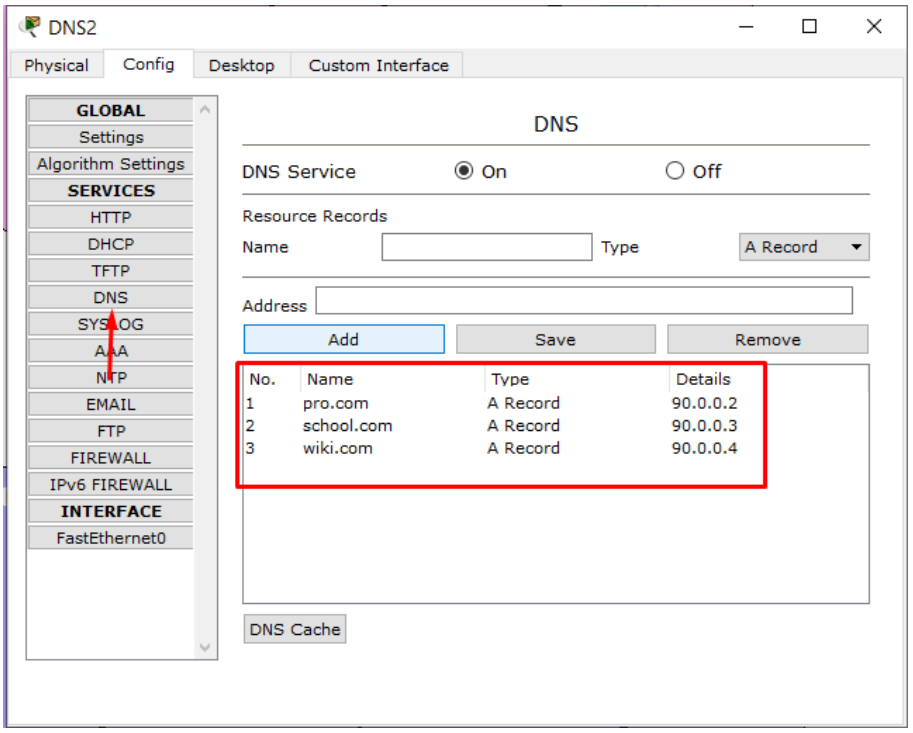


Fig 2: External DNS server setup

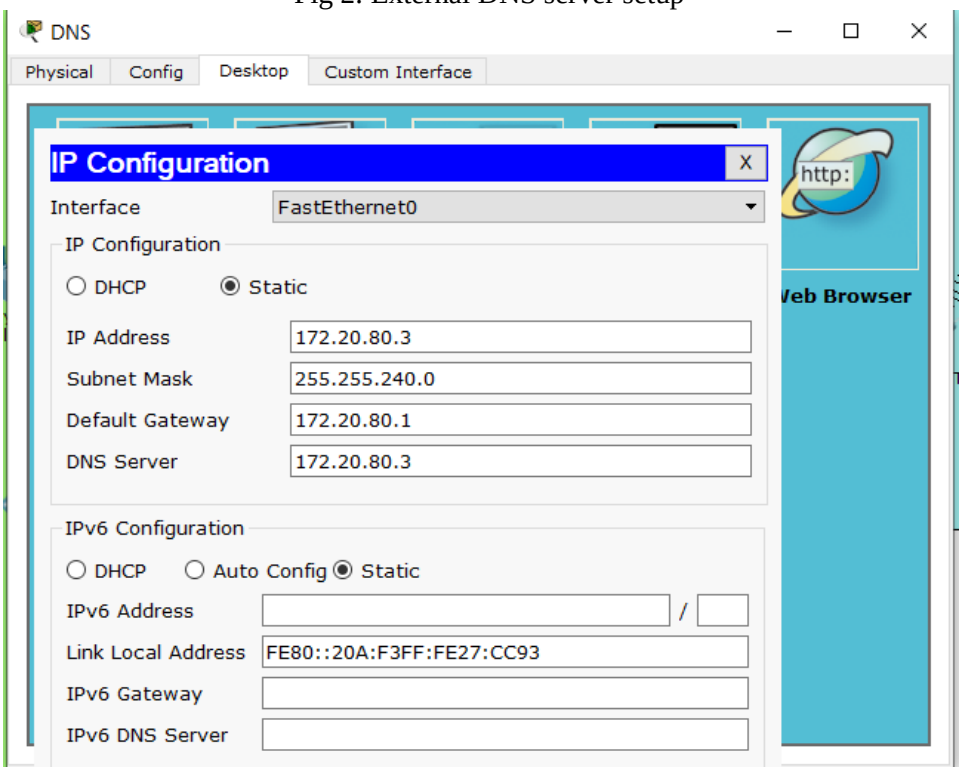


Fig 3: Internal DNS server

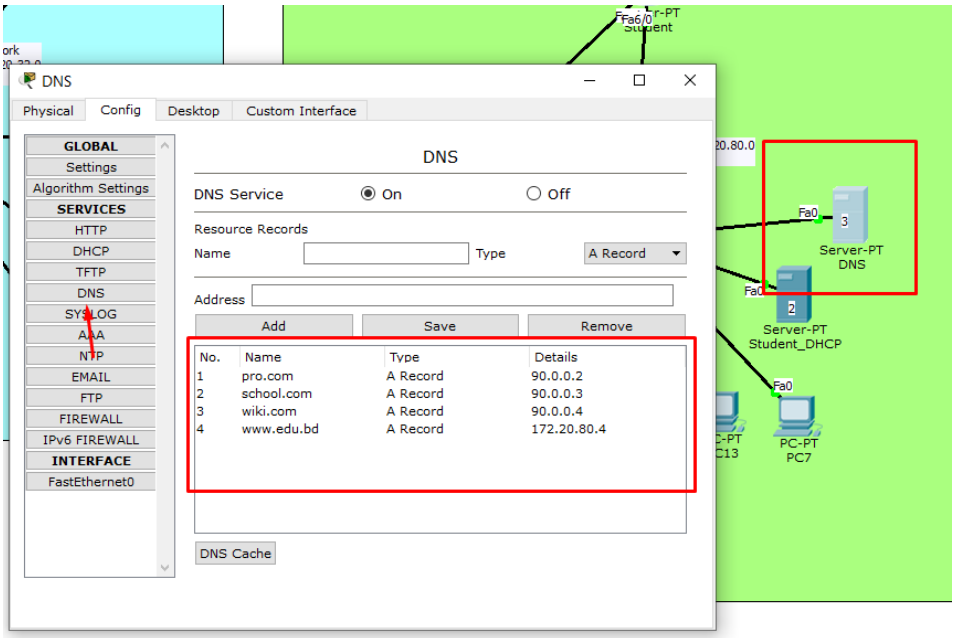


Fig 4: Internal DNS server (with External DNS Cache)

### 3.1.6 HTTPs server setup

When you enter a website's URL into your browser, it sends an HTTP request to the web server hosting that website, which then sends back the web page you requested, allowing you to view it in your browser.
There is total five https servers; 3 external and 1 internal. User can easily use all of them.

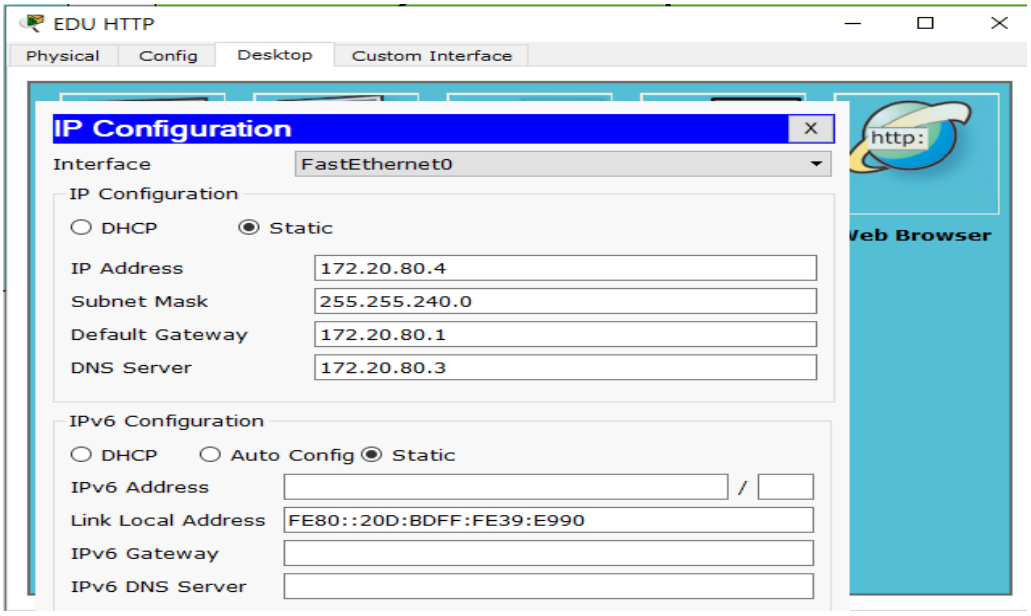


Fig 1: Internal Http server configeration

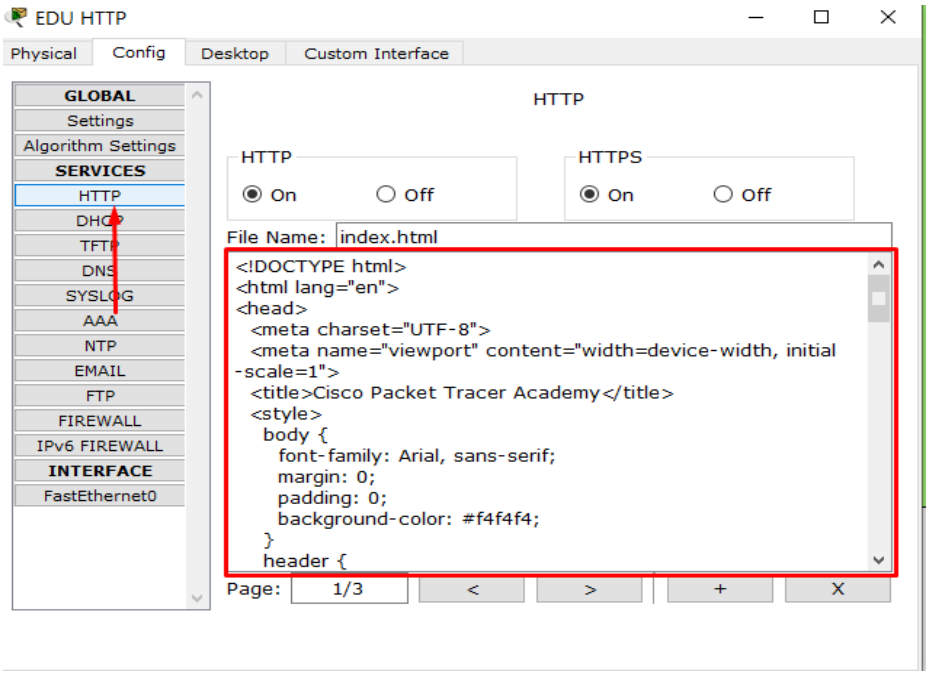


Fig 2: Http server of edu webpage

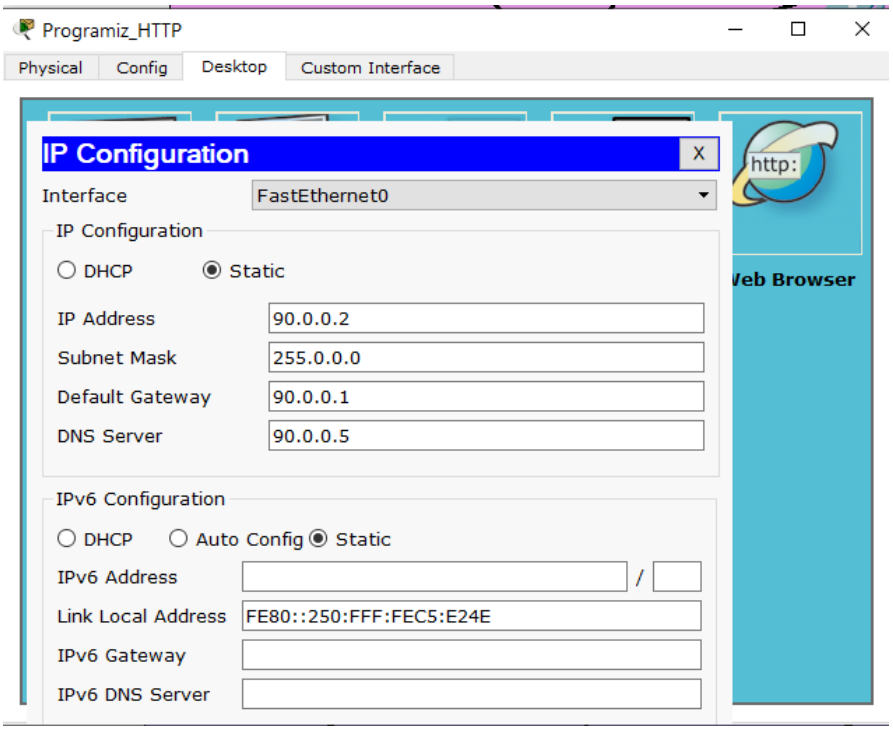


Fig 3: External Http server of configuration

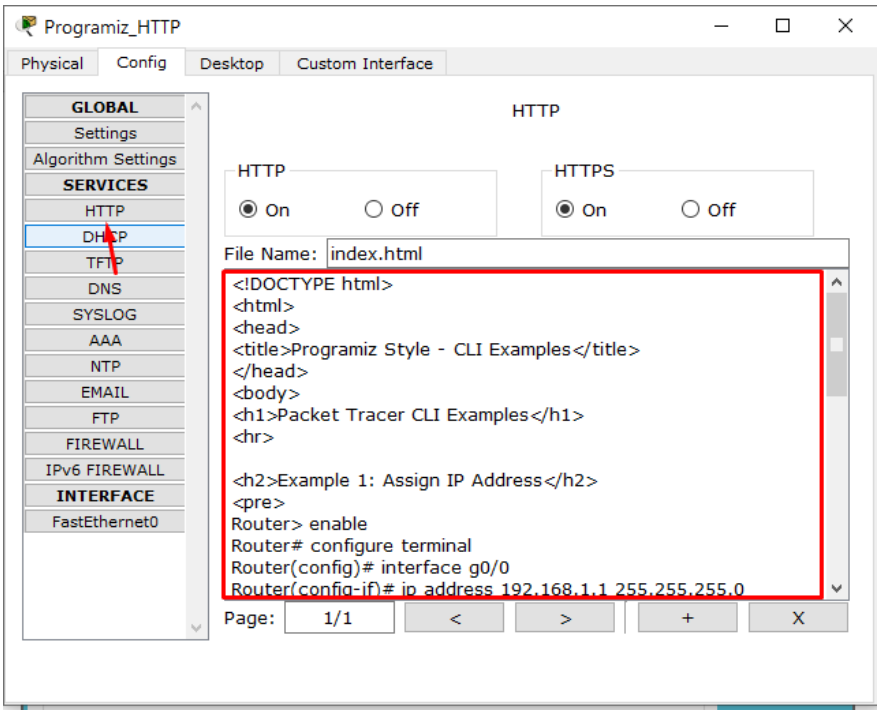


Fig 4: Http server of Programiz webpage

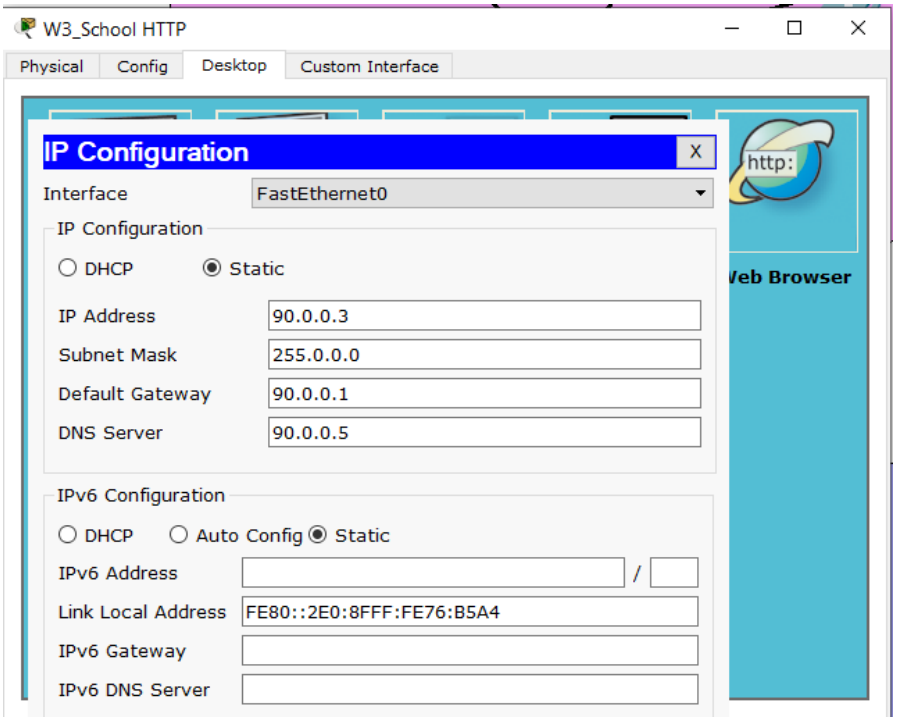


Fig 5: Http server of W3_School webpage

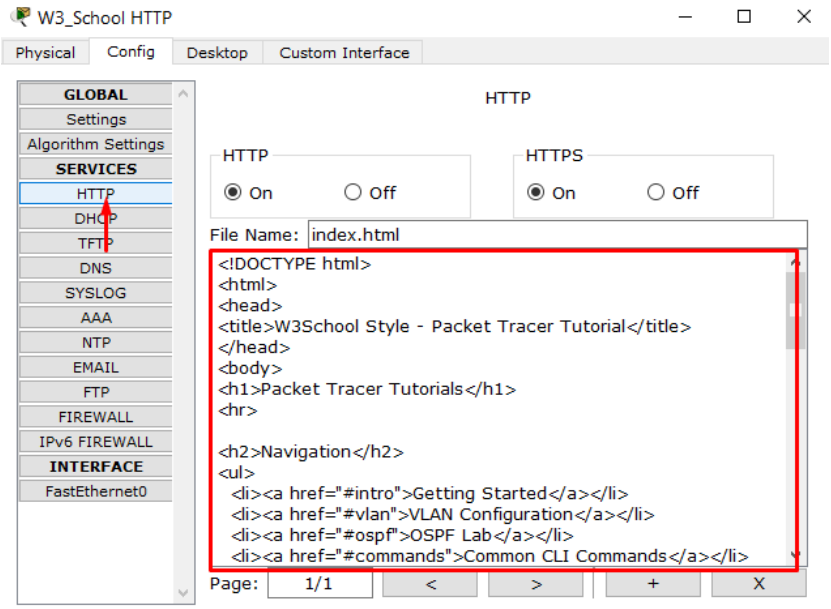


Fig 5: Http server of W3_School webpage

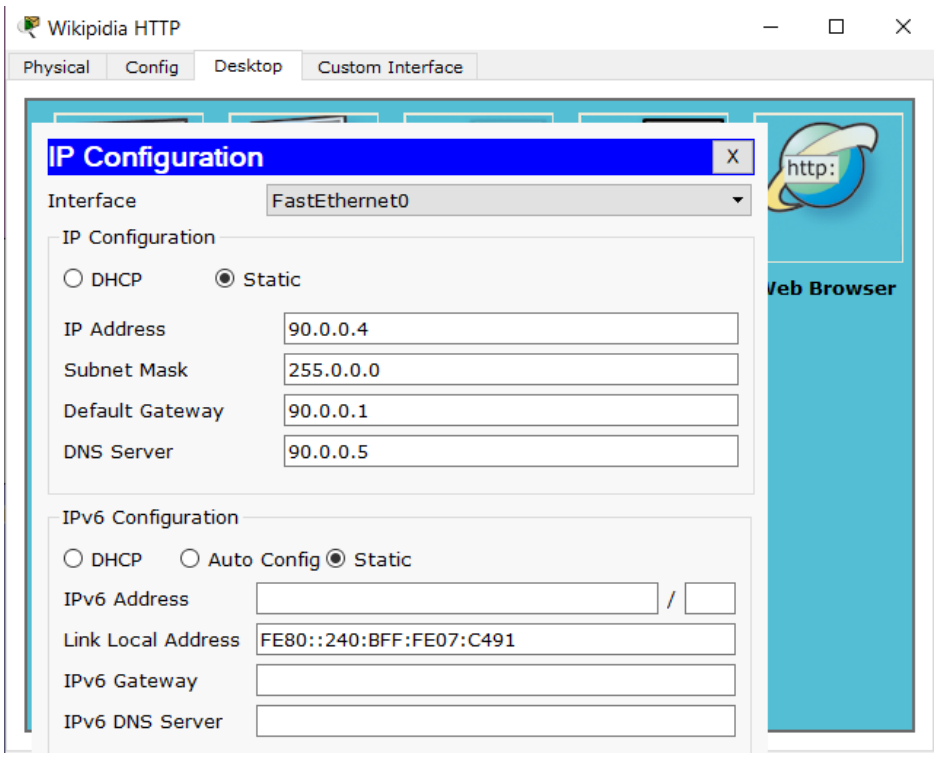


Fig 6: Http server of Wikipidia webpage

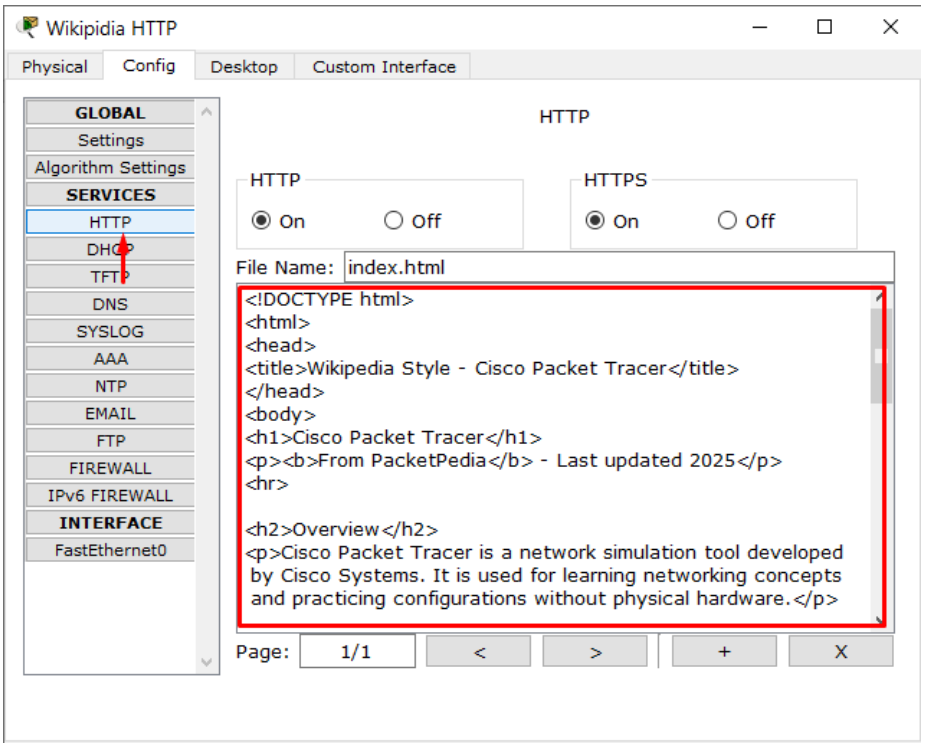


Fig 7: Http server of Wikipidia webpage

### 3.1.7 NAT Configuration

In our School router we have implemented dynamic NAT.

**NAT Configuration for School Router**

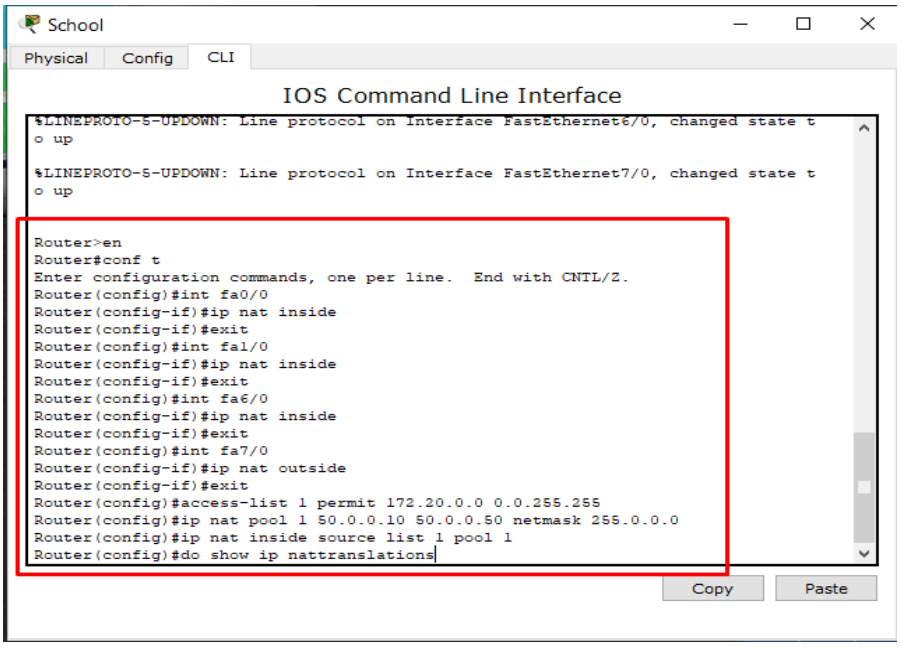


Fig 1: Nat CLI command for School router

## 3.2 Performance Analysis

- **Subnetting Efficiency:**
  - By segmenting the network into Staff, Student, and Guest subnets, traffic is more manageable .
  - Reduced broadcast domains improved network speed and responsiveness.
- **Routing Control**:
  - Static routing provided predictable and secure communication paths.
  - Controlled communication improved system reliability and simplified debugging.
- **Dynamic IP Assignment (DHCP):**
  - Automated IP allocation reduced manual errors.
  - Faster device onboarding and easier management of IP ranges.

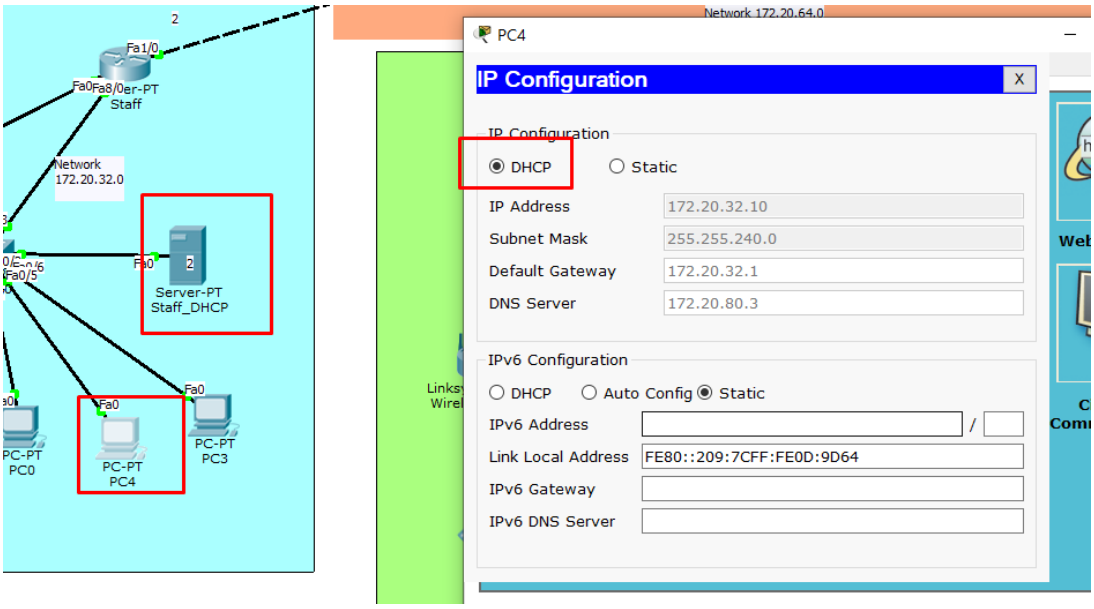


Fig 1: DHCP working properly in End Devices

- **NAT Performance**:
  - Enabled secure, centralized internet access for all subnets.
  - NAT reduced exposure of private IPs, increasing security without performance degradation.
  - Getting reply by using ping in the commands

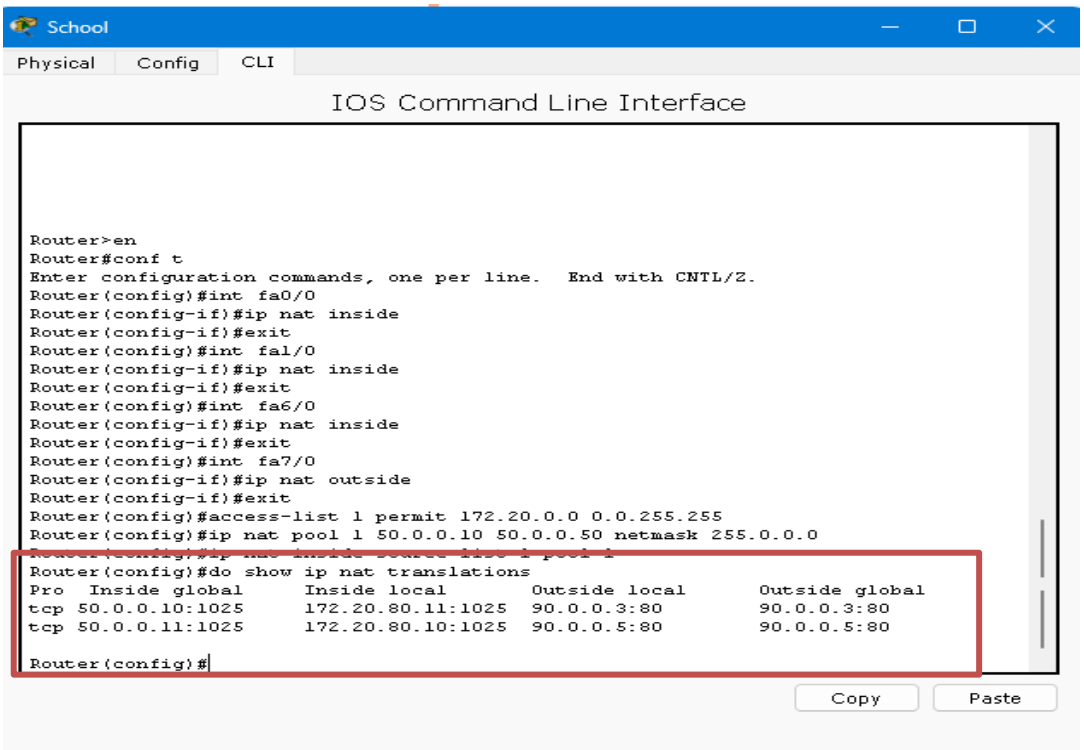


Fig 2: NAT CLI Command line

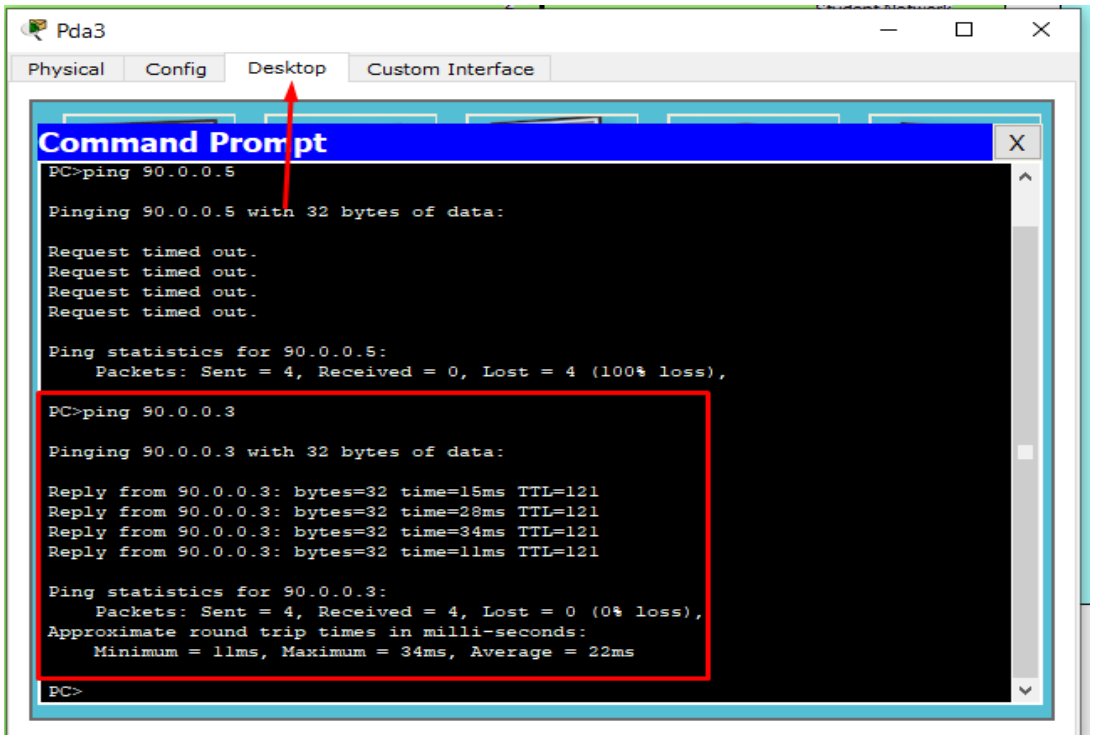


Fig 3: Getting Reply using Ping

- **DNS Responsiveness**:
  - Quick resolution of internal and external domains.
  - Improved user experience due to reduced query latency.

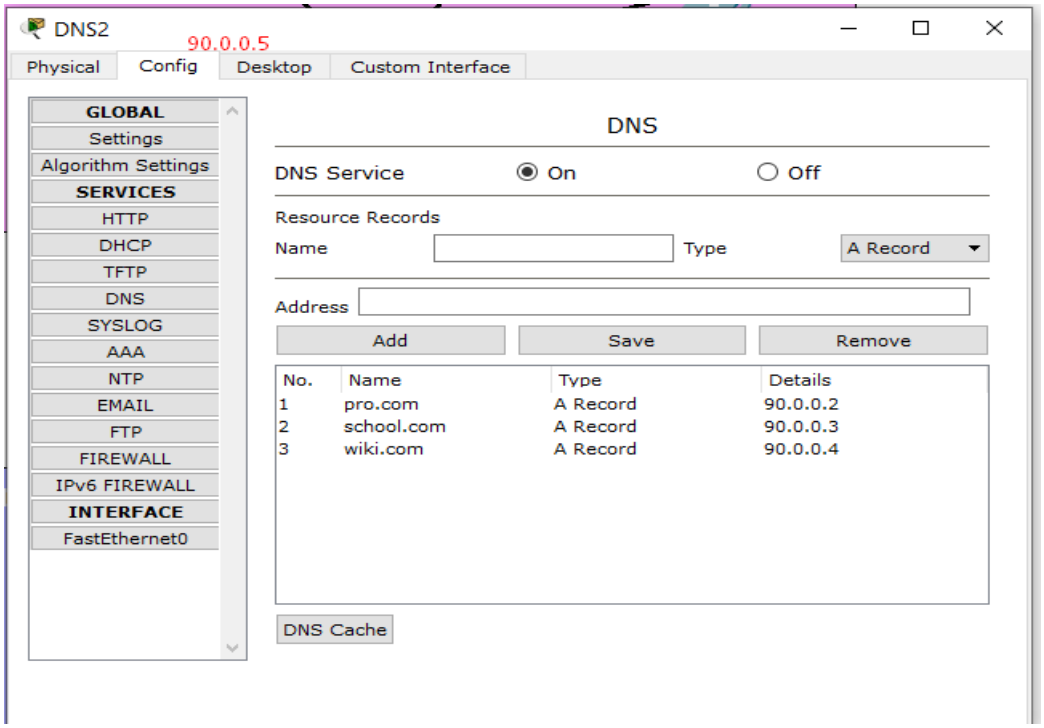


Fig 1: DNS server can be reached through Public IP 90.0.0.5

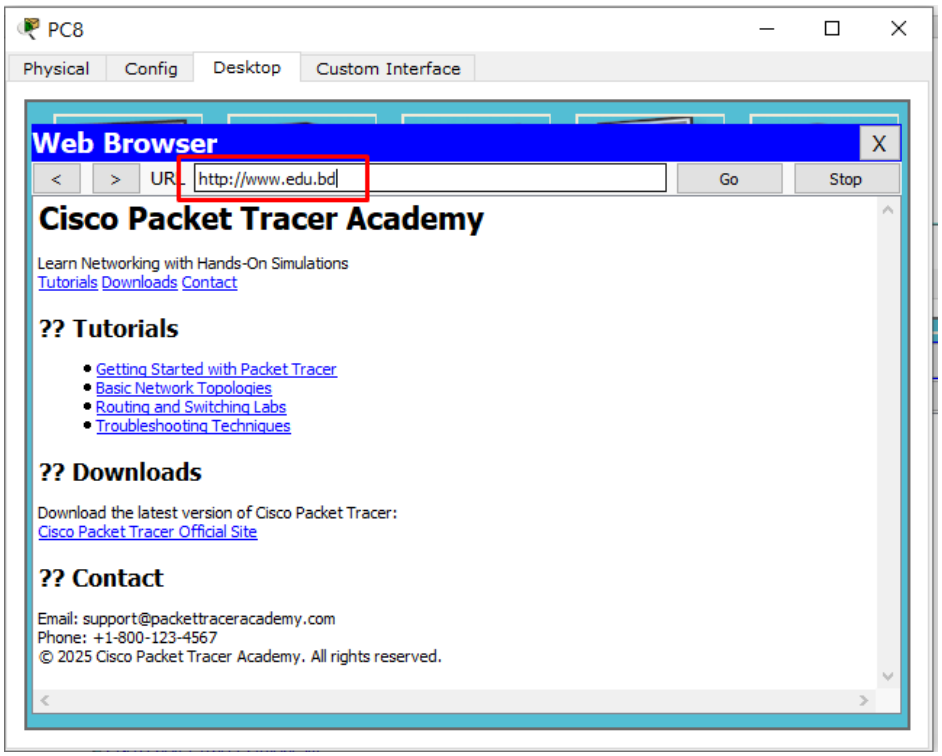


Fig 2: Internal Server **www.edu.bd** can be reached from each subnetted Network

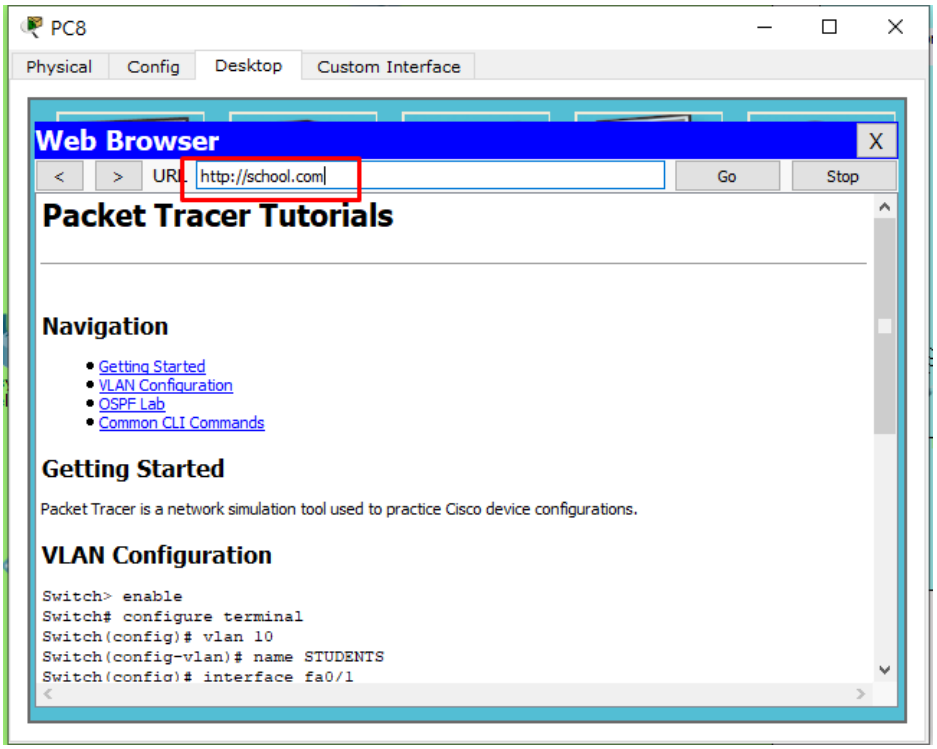


Fig 3: External Server **school.com** can be reached from different network

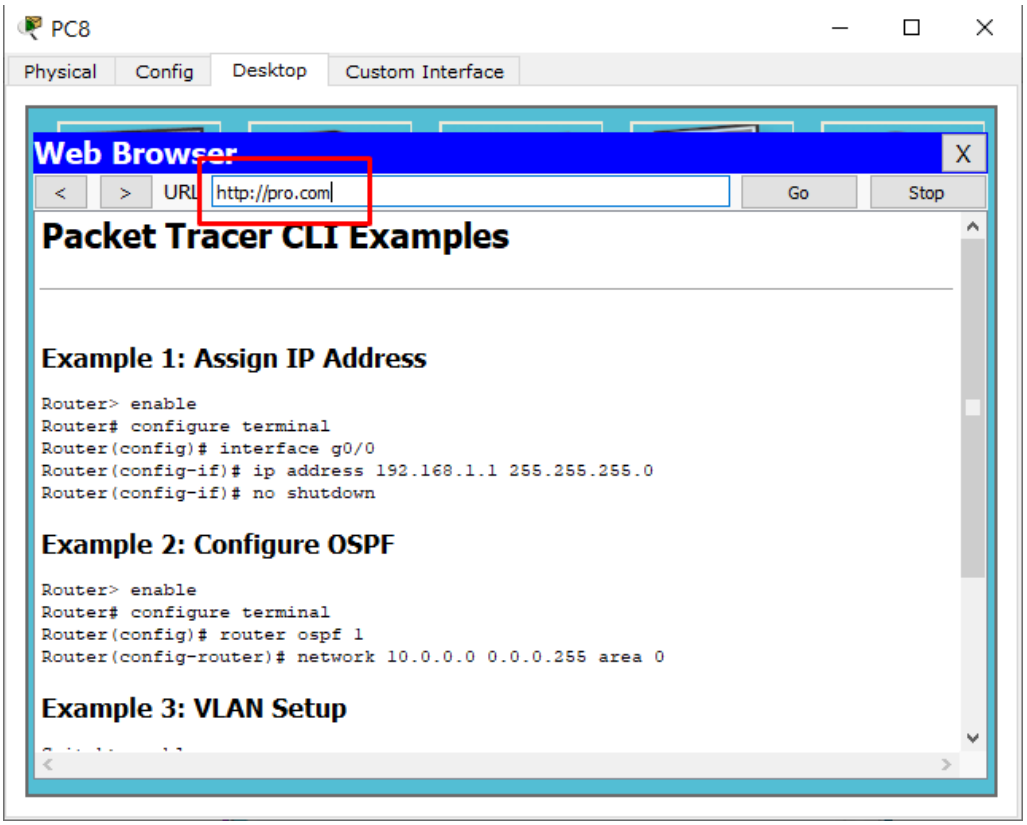


Fig 4: External Server **pro.com** can be reached from every Network

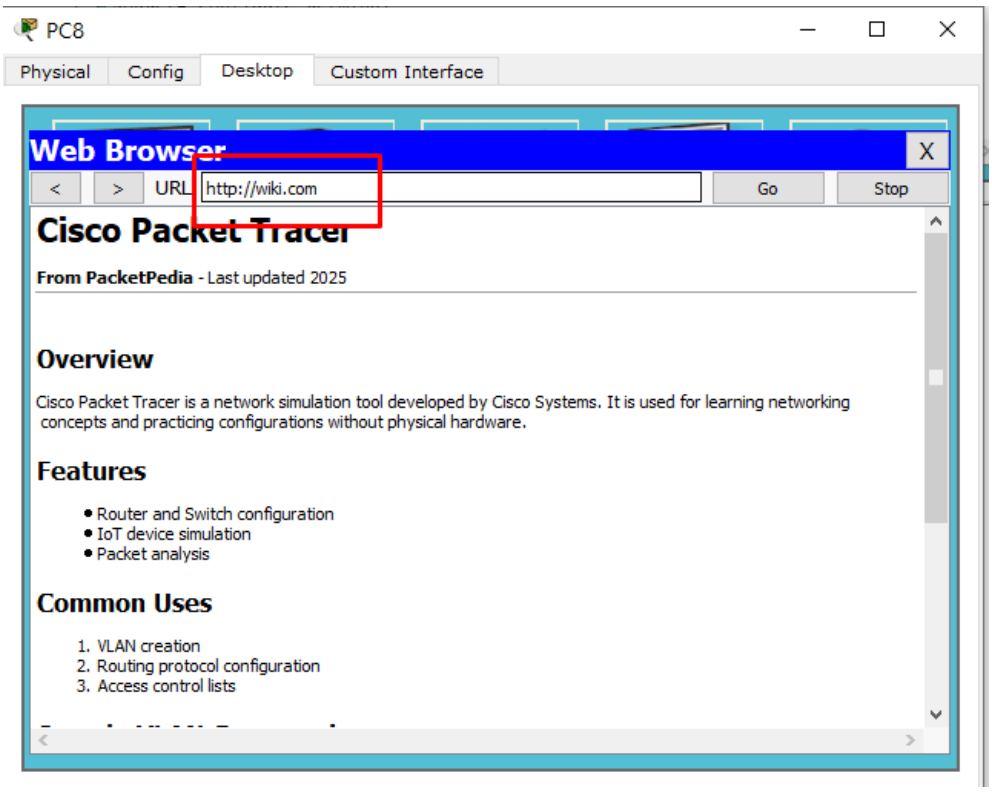


Fig 5: External Server **wiki.com** can be reached from every Network

## Wireless Performance:

- Wireless connectivity was stable and secure.

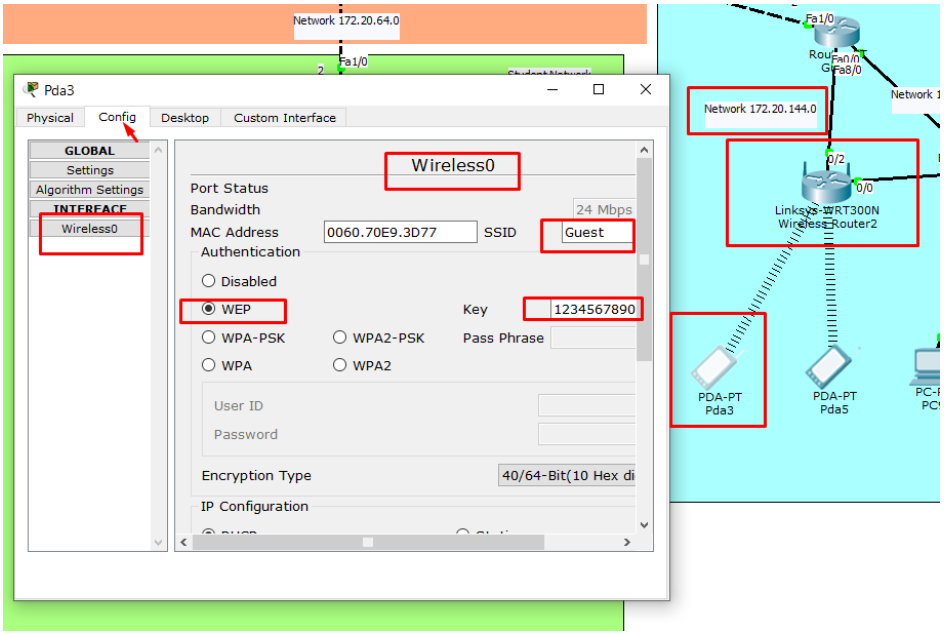


Fig 1: Configuration wireless router

## Overall Network Stability:

- Packet loss was minimal during ping tests.
- All devices maintained consistent connections during simulated usage.

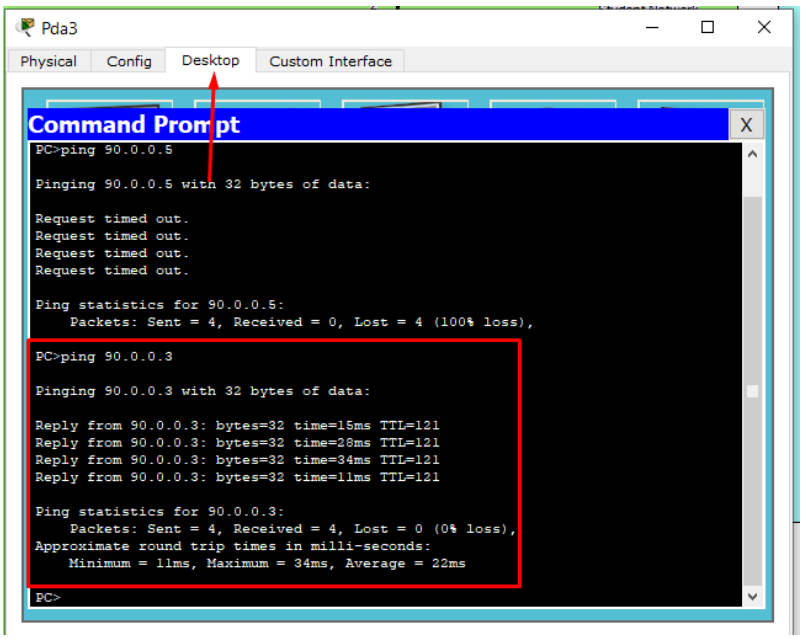


Fig 1: Packet Loss is observed through ping test

## 3.3 Results and Discussion

### Results

The school network with secure internet access was successfully implemented in Cisco Packet Tracer, meeting the design objectives of segmentation, controlled communication, and internet accessibility for staff, students, and guests. The configuration results were verified through multiple connectivity and service tests.

1. All subnets in the private network functioned accurately according to the subnetting plan.
2. All devices in the network received IP addresses through **DHCP**, except for the DHCP server itself, as well as the DNS and HTTP servers, which were configured with static IPs.
3. All devices were statically connected within the private network and could communicate with each other successfully.
4. All devices in the private network could access the internet via **NAT**, except for guest Wi-Fi devices, which were intentionally restricted for security purposes.
5. Default routing was used to simplify the network configuration and ensure that all routers had a straightforward path to reach unknown networks, reducing complexity in managing multiple static routes.
6. The network provided easy access to educational resources such as W3Schools, programming references, and Wikipedia.
7. The **DNS server**, **HTTP server**, NAT configuration, and static routing across the network were all functioning correctly, as confirmed by successful ping tests from end devices in the private network.
8. Wi-Fi routing was implemented for the first time in this setup, using a new approach, and connectivity was stable and functional.

### Discussion

Based on the results, the implemented network design successfully met all technical and functional objectives. The segmentation of staff, student, and guest networks ensured proper access control, reducing the risk of unauthorized data access. Using **DHCP** for most devices minimized configuration errors and made IP management more efficient, while **static IPs** for servers ensured reliability for essential services like DNS and HTTP.

The decision to use **default routing** was strategic—it reduced the complexity of routing tables in a multi-network environment while maintaining reliable communication between network segments. The integration of **NAT** provided secure internet access to internal devices without exposing private IP addresses. Additionally, restricting guest Wi-Fi from internet access in certain cases improved network security and bandwidth management.

Overall, the network was robust, easy to manage, and capable of supporting daily academic and administrative operations. The successful implementation of Wi-Fi routing and stable internet services demonstrated that the network is scalable for future expansions, such as adding more VLANs, integrating firewall rules, or implementing Quality of Service (QoS) for critical applications.

# Chapter 4

# Engineering Standards and Mapping

This chapter ensures that the network meets professional, ethical, and sustainability standards and mapped our work to course outcomes.

## 4.1 Impact on Society, Environment and Sustainability

### 4.1.1 Impact on Life

The school network improves access to educational resources, enabling students, teachers, and staff to work more efficiently. It provides secure internet access for learning, administrative work, and research, enhancing the quality of education and productivity within the institution.

### 4.1.2 Impact on Society & Environment

A well-designed network encourages digital literacy and supports modern teaching methods, such as e-learning and online collaboration. Environmentally, the simulation-based design in **Cisco Packet Tracer** minimizes the need for physical hardware during the planning phase, reducing e-waste and energy consumption.

### 4.1.3 Ethical Aspects

In this networking project, ethical considerations include:

- **Network Security**: Ensuring staff and student data are protected from unauthorized access.
- **Access Control**: Providing internet and internal resources based on user roles without discrimination.
- **Privacy**: Ensuring DNS, DHCP, and HTTP configurations do not expose sensitive user information.
- **Responsible Usage**: Configuring guest access to prevent abuse of resources while maintaining fair usage policies.

By adhering to ethical networking practices, the project promotes trust, security, and compliance with professional IT standards.

### 4.1.4 Sustainability Plan

The network is designed to be scalable and adaptable. As the school grows, new devices and subnets can be added without overhauling the entire system. Virtual simulations for training ensure minimal physical resource usage, making the solution cost-effective and environmentally sustainable.

## 4.2 Project Management and Team Work

We have divide work in three part.

**1. Planning and Plotting**
**2. Implement in Cisco Packet Tracer and**
**3. Bug finding and Fixing faults**

**Simulated Setup:**

- Cisco Packet Tracer version 6.1

**Real-World Setup Budget:**
We make it in Cisco Packet Tracer, so we don't have implemented any real life device so do cost needed. Just a device(PC,Laptop) that supports Cisco Packet Tracer.

## 4.3 Complex Engineering Problem

### 4.3.1 Mapping of Program Outcome

Here we have shown the mapping of the problem and provided solution with targeted Program Outcomes (PO's).

Table 4.1: Justification of Program Outcomes

| PO's | Justification |
|---|---|
| PO1 | Identified and configured network devices (routers, switches, DHCP, DNS) following engineering standards. |
| PO2 | Applied subnetting, address planning, and static routing to manage network traffic effectively. |
| PO3 | Solved real-world network segmentation and access control challenges while maintaining scalability and security. |

### 4.3.2 Complex Problem Solving

In this section, provide a mapping with problem solving categories. For each mapping add subsections to put rationale (Table 4.2).

For P1, you need to put another mapping with knowledge profile and rational thereof.

Table 4.2: Mapping with complex problem solving.

| **EP1** Dept of Knowledge | **EP2** Range of Conflicting Requirements | **EP3** Depth of Analysis | **EP4** Familiarity of Issues | **EP5** Extent of Applicable Codes | **EP6** Extent Of Stakeholder Involvement | **EP7** Inter-dependence |
|---|---|---|---|---|---|---|
| Demonstrated an advanced understanding of routing, NAT, DHCP, DNS, and access control. | Balanced the needs for security and accessibility among different user groups. | Designed detailed subnetting plans and configured access control mechanisms. | Addressed typical school network challenges such as guest isolation and resource access. | Followed best practices and standards recommended by Cisco Networking guidelines. | Addressed the different needs of students, teachers, guests, and administrators. | Integrated DHCP, DNS, NAT, routing, and ACLs into one coherent and efficient system. |

### 4.3.3 Engineering Activities

Here we provided a mapping with engineering activities. For each mapping add subsections to put rationale (Table 4.3).

Table 4.3: Mapping with complex engineering activities.

| **EA1** Range of resources | **EA2** Level of Interaction | **EA3** Innovation | **EA4** Consequences for society and environment | **EA5** Familiarity |
|---|---|---|---|---|
| Used simulation software, virtualized servers, and networking hardware concepts. | Coordinated team roles for subnetting, configuration, and testing. | Combined multiple security techniques like ACLs and NAT to enhance network safety. | Encouraged digital responsibility and sustainability. | Applied knowledge of Cisco networking tools and real-world deployment techniques. |

# Chapter 5

# Conclusion

Here we have summarized the accomplishments of the project, acknowledges its limitations, and suggests directions for future improvement

## 5.1 Summary

This project successfully **designed, configured, and simulated** a secure and segmented **school network** using Cisco Packet Tracer. The network architecture was built around three primary user groups — **Staff, Students, and Guests** — each assigned to its own dedicated subnet for security and efficient traffic management.

The implementation incorporated key networking services and configurations, including:

- **Static Routing** for controlled inter-subnet communication and predictable traffic flow.
- **DHCP** for dynamic IP address allocation, reducing manual configuration errors and simplifying device onboarding.
- **DNS** for both internal and external domain resolution, improving accessibility and usability.
- **NAT** (Static and Dynamic) for secure internet access from internal networks.
- **Multiple HTTP Servers** to simulate internal school web services and external educational sites.
- **Wireless Access Points with Separate SSIDs** for Students and Guests, ensuring role-based access control.

The design proved **scalable, secure, and easy to manage**, making it a strong foundation for real-world deployment in educational institutions. Performance testing confirmed **low latency, minimal packet loss, and stable connectivity**, even under simulated load.

## 5.2 Limitation

- The limitations of this project are primarily related to the constraints of the simulation environment, wireless emulation, and our current level of networking knowledge:
- **Simulation Environment Constraints**
- The Cisco Packet Tracer simulation environment cannot fully replicate the performance of physical networking hardware, such as processing speed, actual throughput, and latency.
- Certain real-world conditions, like hardware failures, environmental interference, and power issues, cannot be accurately modeled.
- The version of Packet Tracer used may lack newer networking features found in modern devices.
- **Wireless Network Limitations**
- Wireless signal range, interference, and bandwidth variations could not be tested in real-world conditions.
- The simulation does not account for obstacles or environmental effects on Wi-Fi performance.
- **Troubleshooting and Bug Fixing**
- Occasional simulation glitches required repeated configurations to fix issues such as device non-responsiveness.

- Certain commands, such as repeated ping tests, sometimes gave inconsistent replies within the simulation.
- **DNS and Host Resolution**.Hostname resolution worked in simulation but was not tested with large-scale domain environments.
- DNS caching behavior in Packet Tracer is simplified and does not fully represent real-world DNS server performance.
- **Knowledge Limitations**
- As this project was completed during the learning phase, some advanced techniques (e.g., redundancy protocols, firewall integration, QoS optimization) were not implemented.

## 5.3 New Learnings

While working on this project, I gained valuable networking knowledge and identified several new concepts that I had to learn in order to make the simulation fully functional:

- **Default Routing** – Learned how to configure default routes so that devices can reach unknown networks through a single gateway, simplifying routing tables.
- **Configuring a Router in a New Approach** – Practiced different router configuration methods, including using CLI commands and exploring alternative topology setups for better network efficiency.
- **DNS Cache** – Understood how DNS caching works to speed up domain name resolution and reduce repeated queries, as well as how to test cached entries in a simulated environment.

These new learnings improved both my technical understanding and practical skills, enabling me to design and troubleshoot the network more effectively.

## 5.4 Future Work

To enhance the school network simulation and address the current limitations, the following improvements can be considered in the future:

1. **Add VLANs for Better Segmentation**
    - Implement VLANs along with subnetting to provide an extra layer of traffic isolation and management.
2. **Configure Access Control Lists (ACLs) More Precisely**
    - Create more detailed ACL rules to control which devices or services can be accessed from each subnet.
3. **Integrate a Backup Router in Simulation**
    - Add a second router in Packet Tracer to simulate basic redundancy and test failover scenarios.
4. **Improve Wireless Security Settings**
    - Experiment with WPA3 (if supported in the simulator) or simulate MAC address filtering for added protection.
5. **Add More Internal Services**
    - Host additional internal services like an FTP server for file sharing or a mail server for communication practice.
6. **Practice with Packet Tracer's Simulation Mode**
    - Use the simulation view to monitor packet flow, troubleshoot connectivity issues, and better understand how data travels in the network.

# References


- Class lecture notes on Computer Networking – [Daffodil International Univarsity], [CSE 321], [Partha Dip Sarkar], [2025].

- Laboratory sessions on Cisco Packet Tracer configuration – [Daffodil International Univarsity], [CSE 321], [Partha Dip Sarkar], [2025].

- *CSExplain* – "Cisco Packet Tracer Tutorials." YouTube, [(199) CSExplained - YouTube].

- OpenAI – ChatGPT (GPT-5) interactive assistance for networking design and documentation.

- DeepSeek AI – Networking guidance and troubleshooting support.

- Diagram AI – Network topology and diagram creation tool.